24 Vital Pulp Therapy

GEORGE BOGEN, TILL DAMMASCHKE, and NICHOLAS CHANDLER

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"There is nothing permanent except change."

—Heraclitus

Vital pulp therapy aims to preserve and maintain pulpal health in teeth challenged by trauma, caries, restorative procedures, and anatomic anomalies. The treatment can be completed for permanent teeth that show reversible pulpal injuries, and outcomes depend on several factors.[87,192,252] The primary objective is to initiate the formation of new mineralized tissue. This procedure is essential for the preservation of involved immature permanent teeth where root development is incomplete, and maintenance of arch integrity is critical during maxillofacial development.[73]

From the first description of direct pulp capping using gold foil by Philipp Pfaff in 1756, dentistry has recognized the ability of the injured pulp to repair.[308] New materials now provide superior sealing properties to protect the pulp from microorganisms and their toxic by-products. Mineral trioxide aggregate (MTA) and other hydraulic calcium silicate cements (CSCs) partnered with advanced treatment strategies have changed the long-held concept that pulp capping after carious exposure should be avoided. The perception that outcomes for direct pulp capping in a carious field are inconsistent is based on traditional protocols and materials that did not generate a favorable environment for hard tissue formation.[168,229,271,382] This has encouraged clinicians towards alternative treatments, such as pulpotomy or pulpectomy, particularly in immature permanent teeth.[74,396] This is further complicated by diagnostic difficulties, because clinical signs, symptoms, and radiographs may not accurately reflect the histologic condition of the pulp (Fig. 24.1).[196] However, based on a better understanding of pulp physiology, caries microbiology, and the inflammatory mechanisms responsible for irreversible changes, teeth with the potential for repair and continued vitality can now be more readily identified and predictably treated.[54]

Maintaining pulp vitality is in contrast to root canal treatment where remaining pulp tissue is removed and the canal system shaped, disinfected, and obturated using root canal filling materials. Although success rates of over 90% can be achieved after approximately 5 years with careful treatment after vital extirpation the complete loss of function of the pulp tissue occurs, which can be disadvantageous.[21] Proprioceptive protection is partially lost after orthograde treatment, permitting more than double the normal occlusal load before a provoked neuromuscular response occurs.[32,314] This proprioceptive deficit can threaten long-term tooth function and survival. In contrast, vital tooth therapy provides a conservative option that exploits bioactive materials and strategies to save the dentition.

This chapter describes the microbiology of caries and the associated physiologic reactivity of pulp tissue. Understanding that pulp tissue has an innate potential for repair in the absence of bacterial contamination, new treatments in vital pulp therapy for the preservation of the pulpally involved permanent tooth are discussed.[97,201]



Fig. 24.1 Radiographs of carious molars in patients aged 12 to 38 years. **A,** Mandibular left first molar in a 23-year-old with minor symptoms. **B,** Asymptomatic maxillary right second molar in a 16-year-old. **C,** Asymptomatic maxillary left first molar in a 38-year-old. **D,** Deep caries in the mandibular right first molar of a 12-year-old. All patients were referred to the endodontist for root canal treatment based on radiographic appearance. They exhibited normal vitality with cold testing, and all were treated successfully with vital pulp therapy. (© Dr. George Bogen.)

The Living Pulp

Vital pulp tissue comprises cells including fibroblasts, undifferentiated mesenchymal cells, odontoblasts, macrophages, dendritic cells, and other immunocompetent cells. The cells of the subodontoblastic layer and odontoblasts form a thin border between the dentin and the periphery of the pulp known as the *Höhl cell layer*.[106,158] The odontoblasts are elongated columnar-shaped cells separated from the mineralized dentin by predentin and characterized by processes that extend into the dentin and possibly to the dentinoenamel junction.[167,348,372,403] Odontoblasts form the mineralized predentin-dentin matrix, which includes phosphoproteins, glycoproteins, proteoglycans, and sialoproteins.[220] Repair mechanisms in the pulp are similar to those in normal connective tissue injured by trauma. When the enamel and dentin are challenged and the pulp is exposed to advancing microorganisms, inflammatory changes can induce pulp necrosis, which precedes problems including infection and its complications.[50,61] The pulp undergoes physiologic, pathologic, and defensive changes during its life.[240,276,381] These include continued dentin apposition, causing reduction of coronal pulp volume and circumference and canal lumen dimensions.[125,277] Atrophy results in fibrosis, dystrophic calcification, degeneration of odontoblasts, and increased cellular apoptosis.[46,125] Aging of human pulp cells is primarily characterized by the formation of reactive oxygen species and senescence-related beta (β)-galactosidase activity.[233] Pain sensitivity is reduced due to a decrease in fast-conducting A-δ fibers and diminished pulp repair, partly attributed to decreases in the levels of substances such as alkaline phosphatase.[254,345]

A comparison analysis of gene expression levels reflecting cell function, proliferation, differentiation, and development found them markedly higher in young pulps.[381] Young dental pulps show greater cell and tissue differentiation, proliferation, and development of the lymphatic, hematologic, and immune systems compared to older pulps where the apoptosis pathway is highly expressed.[381] Although A-β fiber function remains constant with aging, a decrease in A-δ fibers with age may reduce the perception of pain by these fibers.[254]

In summary, the main functions of the dental pulp include dentin formation during development and life of the tooth, the transmission of stimuli via proprio- and pain receptors, and immune responses. The pulp also produces reparative dentin as a defense mechanism against external stimuli, and tissue during the formation and closure of the root apex.

Loss of Pulp Vitality

Pulpless teeth with minimal remaining tooth structure that undergo root-canal-filling procedures and are heavily restored become more vulnerable to irreparable fracture because of the loss of proprioceptive mechanisms.[32,262,314,358] Moisture depletion from dentin and reduction of tooth stiffness are minimal after root canal treatment.[186,204,241] Although root canal treatment can prolong tooth survival, the cumulative loss of tooth structure from endodontic and restorative treatment weakens teeth.[64,76,118,393] Changes in root canal geometry are inevitable in the course of canal preparation and may lead to a higher incidence of fractures.[151,235]

Other potential problems are tooth discoloration and increased caries risk because of plaque accumulation and altered microflora or due to the absence of the defenses of the pulp-dentin complex.[224,263] Root-canal-filled teeth demonstrate an increased susceptibility to recurrent caries because of poor marginal integrity of restorations or modification of their biologic environment.[78,263] Root canal treatments may be more complex than initially thought, while procedures for maintaining vitality are conservative, easy to implement, and often cost-effective.[198,339]

Pulpal Response to Caries[a]

The microbial advance in carious lesions is the primary cause of pulpal inflammation and potential necrosis. Acidogenic gram-positive bacteria, predominately oral streptococci and lactobacilli, produce lactic acid as the main metabolic by-product in active lesions that demineralizes enamel and dentin.[136,176] Immune responses and inflammation occur when the caries front is within 1.5 mm of the pulp and bacterial antigens and metabolites diffuse through dentinal tubules.[43,136,203,293] If bacterial challenge continues, immune cell responses lead to increased inflammation and edema, initially characterized clinically by pulpal pain. In the low-compliance environment of the pulp space, inflammation eventually leads to pulp disintegration and apical pathosis.[176,376]

Advancing caries is initially confronted by protective innate immune responses, progressing to an adaptive response when bacteria directly approach the pulp.[175] The pioneer microbes first encounter a positive outward flow of dentinal fluid, characterized by the deposition of immunoglobulins and serum proteins that slow the diffusion of antigens.[176] Potent microbial metabolites, such as lipoteichoic acid and lipopolysaccharide, also activate the innate immune system. The by-products stimulate signaling by Toll-like receptors in odontoblasts when these cells first encounter the carious front. They also stimulate proinflammatory cytokines, including interleukin-1, interleukin-8, interleukin-12, tumor necrosis factor alpha, vascular endothelial growth factor (VEGF), and transforming growth factor beta (TGF-β).[137,187,411] VEGF promotes vascular permeability and angiogenesis. Dentin mineralization and matrix metalloproteinase secretion are also induced by the increased expression of TGF-β. Cariogenic bacteria also activate complement pathways and induce the proinflammatory cytokine interferon γ, responsible for killing phagocytosed bacteria by activated macrophages.[190] Odontoblasts also participate in the adaptive response by the synthesis of reactionary dentin, which features diminished tubularity.[86] As microorganisms advance, the modified helical structure in reparative dentin constricts lumen diameter, forming a barrier against pathogens.[85]

As pulpitis progresses, vasoactive neuropeptides contribute to increased vascular permeability and intrapulpal blood flow. Increases in neuropeptide concentration and nerve sprouting characterize neurogenic inflammation, which can cause a transient increase in interstitial tissue pressure and contribute to painful pulpitis.[65,323] Immune cells attempt to control neurogenic inflammation with the secretion of peptides such as somatostatin and β-endorphin.[280] The primary effector cells in innate responses include natural killer cells, neutrophils, monocytes, and macrophages. Immature dendritic cells and T cells contribute to immunosurveillance during the progression of caries. Macrophages eliminate both pathogens and senescent cells while contributing to tissue homeostasis by repairing and remodeling tissue after inflammation.[175]

Cytokines are small, cell-signaling proteins secreted by innate immune cells that induce phagocyte extravasation during inflammation. Chemokines secreted by odontoblasts, fibroblasts, immature dendritic cells, and macrophages stimulate leukocyte recruitment by directing monocyte and neutrophil migration extravascularly to sites of infection.[175] Prompt intervention, through removal of caries and bacterial antigens before irreversible pulpitis commences, can resolve inflammation (Fig. 24.2).

Dentin Regeneration and Reparative Tissue Formation

Mammalian dentin can be subdivided into primary, secondary, and tertiary types. Primary dentin is regular, tubular, and develops before eruption. Secondary dentin is also regular and formed throughout life. Tertiary dentin is more or less irregular and forms locally in response to caries or cavity preparation.[228] It is also referred to as irritation dentin. Secondary dentin formation is slower than that of primary dentin as odontoblast activity is presumably downregulated by autophagocytosis after eruption.[95] A mild external stimulus to the odontoblasts leads to upregulated activity and tertiary dentin formation.[350,353] Unlike bone, which is able to adapt to physiological changes throughout life, dentin is neither remodeled nor replaced once lost.[319]

TERTIARY DENTIN FORMATION

Once dentin is exposed by caries, trauma, or cavity preparation, the tubules are opened and odontoblasts injured or even destroyed. If these challenges are of moderate intensity, they lead to a brief inflammation, which subsides when the irritant is removed. Thereafter, tertiary dentin is formed in the affected region.[319,353] Minor trauma to the pulp stimulates undamaged primary odontoblasts to form new dentin.[236,353,354] This tertiary dentin is therefore also referred to as a reactionary dentin, defined as dentin whose matrix is secreted by primary postmitotic odontoblasts in response to an adequate stimulus.[236,353,354]

Injury to the dentin and odontoblast processes leads to a release of growth factors, which in turn leads to the upregulation of odontoblast activity.[350,353] The probable control mechanism includes a series of tissue factors or signaling molecules, mainly from the TGF-β family released from the dentin matrix following injury.[156,157] These growth factors can also be secreted by odontoblasts, promoted by acidogenic bacteria, or removed from the dentin by different rinsing solutions used during treatment such as ethylenediaminetetraacetic acid (EDTA), sodium hypochlorite (NaOCl), or citric acid.[352,353] In addition, the application of calcium hydroxide (CH) or silicate cements (e.g., MTA or Biodentine) to dentin or

[a]Adapted from Hahn and Liewehr.[175–177]

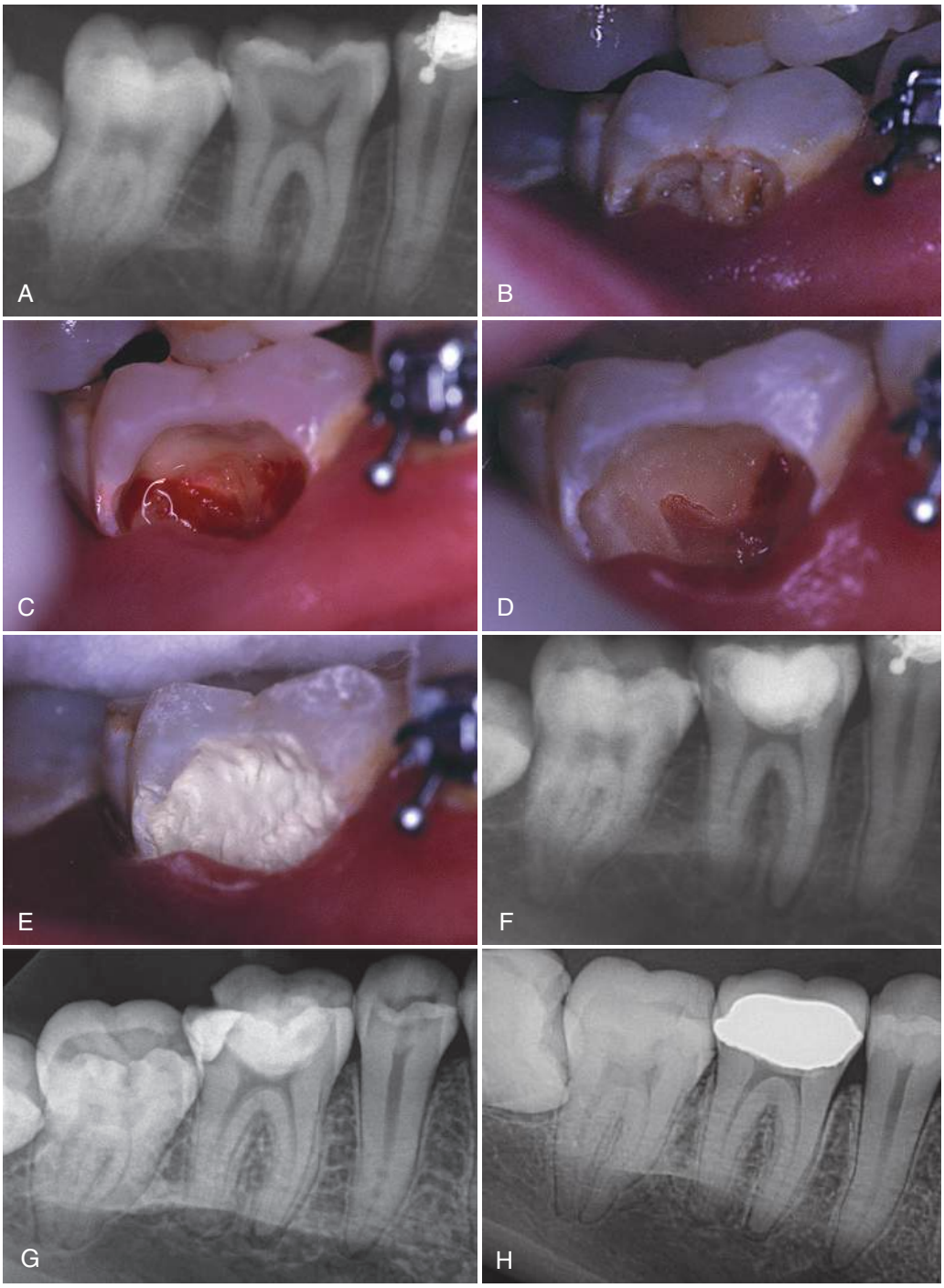


Fig. 24.2 A 15-year-old female patient presented with full orthodontic banding and deep cervical caries of the mandibular right first molar (tooth #30). The tooth was sensitive to cold and biting pressure. **A,** Periapical radiograph revealing large carious lesion and normal periapical structures. **B,** Clinical photograph showing advanced cervical caries extending below crestal gingiva. **C,** View of initial pulp exposure during caries excavation guided by caries detector dye. **D,** Photograph after complete caries removal, large pulp exposure and 6% sodium hypochlorite hemostasis. **E,** Clinical photograph after placement of thick white mineral trioxide aggregate aliquot over entire pulp and surrounding dentin. **F,** Radiograph showing pulp capped molar with temporary restoration (Photocore). **G,** Four-year radiographic control showing recurrent caries and fractured distal marginal ridge tooth #30. **H,** Five-year radiographic review showing no detectable apical pathosis and recently cemented full coverage porcelain fused to metal crown. The molar was asymptomatic and responded normally to sensibility testing. (© Dr. George Bogen 2019.)

onto pulp tissue leads to TGF-β1 release.[164,232,374] Residual dentin thickness has a significant influence on dentin formation. Even if the dentin above the pulp remains intact, not all odontoblasts will survive. In cavities with a residual dentin thickness of less than 0.25 mm, approximately 50% of cells survive; in shallower cavities, this figure is 85% or more.[281] Despite odontoblast death, hard tissue formation can be observed. The reactionary dentin formation by the primary, postmitotic odontoblasts under these conditions is therefore a specific form of tertiary dentin.

MINERALIZED TISSUE FORMATION AFTER LOSS OF THE PRIMARY ODONTOBLASTS

The formation of mineralized tissue after loss of the odontoblasts is more complex than that of reactionary dentin. Dentin formation is not possible without odontoblasts. Nevertheless, after direct capping of a healthy, noninflamed pulp, hard tissue formation occurs. This mineralized tissue is heterogeneous, amorphous, and atubular. Therefore it differs histologically from primary and secondary dentin, but also from reactionary dentin. It is questionable whether this tissue can be called a "dentin bridge," as it is not dentin from a histological viewpoint. If it really was further formation of dentin, a new type of cell would have to replace the destroyed primary odontoblast. The origin and differentiation of these initial cells remains unclear.[319]

Recent investigations have found no histological evidence for the formation of replacement odontoblasts or new odontoblast-like cells after direct pulp capping (DPC) of human teeth with either CH or MTA.[111,319,320] Secondary odontoblasts or odontoblast-like cells are therefore histologically not detectable in human teeth. It is uncertain whether the hard tissue formation is really dentin or just a dystrophic intrapulpal mineralization in response to inflammation.[111,320] However, it is important to emphasize that both reactionary dentin (formed by primary odontoblasts) and hard mineralized repair tissue can form simultaneously in the same lesion.[319,353]

Indications for Vital Pulp Therapy

The penetration of microorganisms and their metabolic products through dentin leads to an inflammation in the pulp. Mediated via cell receptors on odontoblasts, dendritic cells, and pulp fibroblasts, an immune response is initiated. This leads to hyperemia; the reaction is characterized by a decrease in cell number, a flattening of the odontoblasts, and the immigration of lymphocytes and plasma cells.[321] Clinically, this correlates with the development of reversible pulpitis, where healing could be facilitated by therapeutic intervention. If the stimulus persists, bacterial colonization in the pulp chamber can be detected, microabscesses and tissue necrosis lined with polymorphonuclear neutrophilic granulocytes become present, and inflammatory infiltrates are found in the periphery.[321] This is considered to be irreversible pulpitis.

Signs of a reversible pulpitis are a positive response to cold sensibility testing with radiating pain. An irreversible pulpitis is diagnosed on the (enhanced) positive sensibility test, on radiating irritant-persistent pain or lingering pain, pain on heat, and possibly on the patient's inability to localize the problem tooth. However, irreversible pulpitis may be asymptomatic.[14] The correct diagnosis of pulp condition before DPC plays an important role in the outcome. Therefore, assessing the possible the degree of inflammation and/or infection is required.[211] It is challenging to assess pulp health accurately via clinical tests and to distinguish between altered and healthy pulps.[11]

A positive sensibility test (cold) can give a variety of different diagnoses, depending on the severity and duration of response, guiding clinicians to different diagnoses. Teeth responding to cold test may lead to a preoperative diagnosis of normal pulp, reversible pulpitis, or irreversible pulpitis dependent on subjective responses.[301] Sensibility testing does not precisely mirror the condition of the pulp, with 10% to 16% of results being false.[19] Besides pain history and sensibility testing, hemorrhage after pulp exposure should be assessed. This may be a more reliable way to determine the status of pulpal infection than clinical signs and symptoms or sensibility testing.[252] Bleeding may reflect the level of inflammation of the pulp, with excessive hemorrhage usually indicating a pulp with a decreased or no chance of recovery.[88] The inflammatory response extends deeper into the pulp tissue when carious dentin is present during exposure, with possible bacterial penetration, compared to superficial inflammation when the pulp is just mechanically exposed.[229]

Pulps with profuse and lingering hemorrhaging had a significantly poorer outcome than those with modest bleeding or bleeding of short duration.[252] Clinically, bleeding should be controlled within 5 to 10 minutes.[211] According to current recommendations in cases of irreversible pulpitis, the healing of the pulp tissue cannot be achieved predictably after removal of the triggering stimulus (e.g., caries). The diagnosis "irreversible pulpitis" often leads to the initiation of pulpectomy and root canal treatment. Although there is evidence that the histological observations described earlier correlate well with the clinical diagnosis, it should be noted that the clinical pattern of the complaint does not indicate the tissue's ability to regenerate.[321] In a paradigm shift, current clinical investigations using CSCs for full pulpotomies in adult teeth diagnosed with irreversible pulpitis and demonstrating radiographic periapical rarefaction have been treated successfully without pulpectomy and conventional orthograde treatments.[22,23,25,211,242,363,364,386] Refer to the section Pulpotomy With Calcium Silicate Cements.

Maintaining pulp vitality can only be successful if bacterial infection of the tissue during and after the therapy is eliminated. Vital pulp treatments should generally be performed on teeth that do not show pronounced and unprovoked pain symptoms or severe pain on percussion. Vital pulp therapy should not be performed without a response to the sensibility test (in which the pulp status must be verified after pulp exposure) alongside radiographic evidence of periapical disease. In cases diagnosed with irreversible pulpitis, advanced treatment options such as CSC pulpotomy should be considered, especially in younger patients. All vital pulp treatment procedures require a quality bacteria-tight coronal restoration to preclude infection, with dental dam placement, the use of sterile instruments and complete excavation of caries. If not, root canal treatment should be given preference or extraction for teeth deemed unrestorable.[107]

Favorable conditions exist in young patients with pulps without previous damage.[334] With increasing age, a reduced ability for repair is expected due to pulpal changes characterized by a cell-poorer and more fibrous pulp tissue.[161,285] Nevertheless, the age of the patient seems to play only a minor role in treatment success.[23,101–103,206,211,242,243,248,258,363] The same applies to factors such as the sex of the patient, tooth position (posterior or anterior), and the size or location of the pulp exposure.[110,206,244,260,301] The success rates of vital pulp therapies in the literature vary widely, especially in

the case of DPC after caries excavation. Early clinical failures (within days or weeks) are multifactorial, but certainly correlate with improper pulpal diagnosis or inability to identify and remove necrotic pulp tissue. An inaccurate pulpal diagnosis can underestimate the inflammatory status of the pulp, leading to irreversible pulpitis and pulp necrosis with postoperative pain.

The introduction of new bioactive materials, along with modified protocols, make more teeth with deep caries, traumatic injuries, and mechanical exposures viable candidates for innovative pulp therapies designed to maintain pulp survival. Outcomes depend on case selection, hemostatic agents, choice of pulp capping material, and the integrity of the sealed permanent restoration. The purpose in vital pulp therapy is to avoid or delay root canal treatment and advanced restorative care because these procedures, together, reduce long-term tooth survival compared to teeth with vital pulps.[76,77,118,270,289,328,366,380,385]

Procedures Generating Mineralized Tissue Barriers

INDIRECT PULP CAPPING

Indirect pulp capping is defined by the American Association for Pediatric Dentistry (AAPD) as "a procedure performed in a tooth with a deep carious lesion approximating the pulp but without signs or symptoms of pulp degeneration. Indirect pulp treatment is indicated in a permanent tooth diagnosed with a normal pulp with no signs or symptoms of pulpitis or with a diagnosis of reversible pulpitis."[13] Hence indirect pulp capping is the permanent capping of a thin, pulp-nigh, cariously altered dentin layer, without complete excavation of the caries.[33] This may differ from the definition in other countries, for example, in the German-language literature, "indirect pulp capping" is defined as the treatment of a thin layer of caries-free dentin remaining above the pulp chamber with a medication.[332] The French definition of indirect pulp capping implies the covering of both healthy or carious dentin.[230] These definitions may lead to confusion when interpreting the literature.

Clinically, this situation usually arises in the excavation of an extensive carious lesion, so that indirect capping is also called *treatment of deep carious lesions*. A more general term would be "treatment of the pulp-nigh dentin," since indirect pulp capping may be necessary even in caries-free teeth, for example, after dental trauma.[357] Because only a minimal dentin layer remains above the pulp tissue, there is a risk of irreversible inflammation of the pulp via the dentinal tubules.[284] This can result from remaining bacteria or microorganisms actively entering the tissue and by cytotoxic components from restorative materials diffusing across thin residual dentin. Pulp-nigh dentin should be disinfected and bacteria-sealed with a capping material, stimulating the formation of tertiary dentin.[319] The indirect capping thus serves to protect the vital pulp, especially after caries removal. If there is already reversible pulpitis, a favorable environment for pulpal healing should be generated by indirect pulp capping (Fig. 24.3).

Advancing microorganisms and their by-products during the carious process pose a threat to the pulp.[322] Therefore, during caries excavation, the number of microorganisms in the cavity and near the pulp should be reduced, with indirect capping completed under dental dam isolation. To prevent spread of microorganisms, it is recommended to disinfect the clinical crown with NaOCl (1% to 5%) or chlorhexidine digluconate (CHX, 2%) prior to excavation. After caries excavation, the cavity should be cleaned with NaOCl or CHX and water spray.[53,75] Damage to vital tissue when using NaOCl is not of concern.[31] Nevertheless, it remains unclear how much altered dentin can be left while allowing the pulp to heal.[50,338] Hence indirect capping materials should eliminate potential residual microorganisms, neutralize any acidic (due to the carious defect) tissues, remineralize dentin, and stimulate the pulp to form tertiary dentin. Due to the disadvantages of aqueous CH suspensions, the use of CSCs can be seen as an improved alternative for indirect capping.[9]

DIRECT PULP CAPPING

Direct pulp capping is defined as "placing a dental material directly on a mechanical or traumatic vital pulp exposure" and "sealing the pulpal wound to facilitate the formation of reparative dentin and maintenance of a vital pulp."[14] The procedure is indicated for pulp exposures incurred as a result of caries removal, trauma, or tooth preparation. When mechanical exposures occur during tooth preparation, the exposed tissue is generally not inflamed. However, in cases of trauma or carious exposure, the degree of inflammation is the key predetermining prognostic factor. According to the American Association of Endodontists (AAE), "In a carious pulp exposure, underlying pulp is inflamed to a varying or unknown extent."[14] The major challenge in DPC is the proper identification and removal of the acutely inflamed or necrotic tissue compromised by longstanding proximity to oral microorganisms.[247] Direct capping is therefore geared towards the healing of the pulp and the preservation of the tooth. If the pulp survives this treatment (remains vital), this can be considered a clinical success. Direct capping is a noninvasive, relatively simple and inexpensive treatment technique that, unlike root canal treatment, often does not require extensive restorative care. [198]

Since Hermann's first publications in the late 1920s, CH has been used as an aqueous suspension for direct capping.[189] However, the published success outcomes of DPC in a carious field show significant variation and the procedure remains controversial. Some recent publications therefore recommend selective excavation and leaving carious tissue in the tooth, suggesting improved performance over direct capping. For a direct comparison of the two treatments, there are currently only three clinical studies available, two comparing 1- and 5-year data and the third showing 1-year data.[25,51,52] In the second study, the prognosis of the two-stage excavation was compared with complete caries excavation treated by DPC or partial pulpotomy.[51] Five years after two-step caries excavation, the pulp was vital in 60% of the cases, whereas in the same period the success rates after direct capping were only 6% and after partial pulpotomy 11%.[51] The reason for lower success rates may include the absence of disinfection after pulp

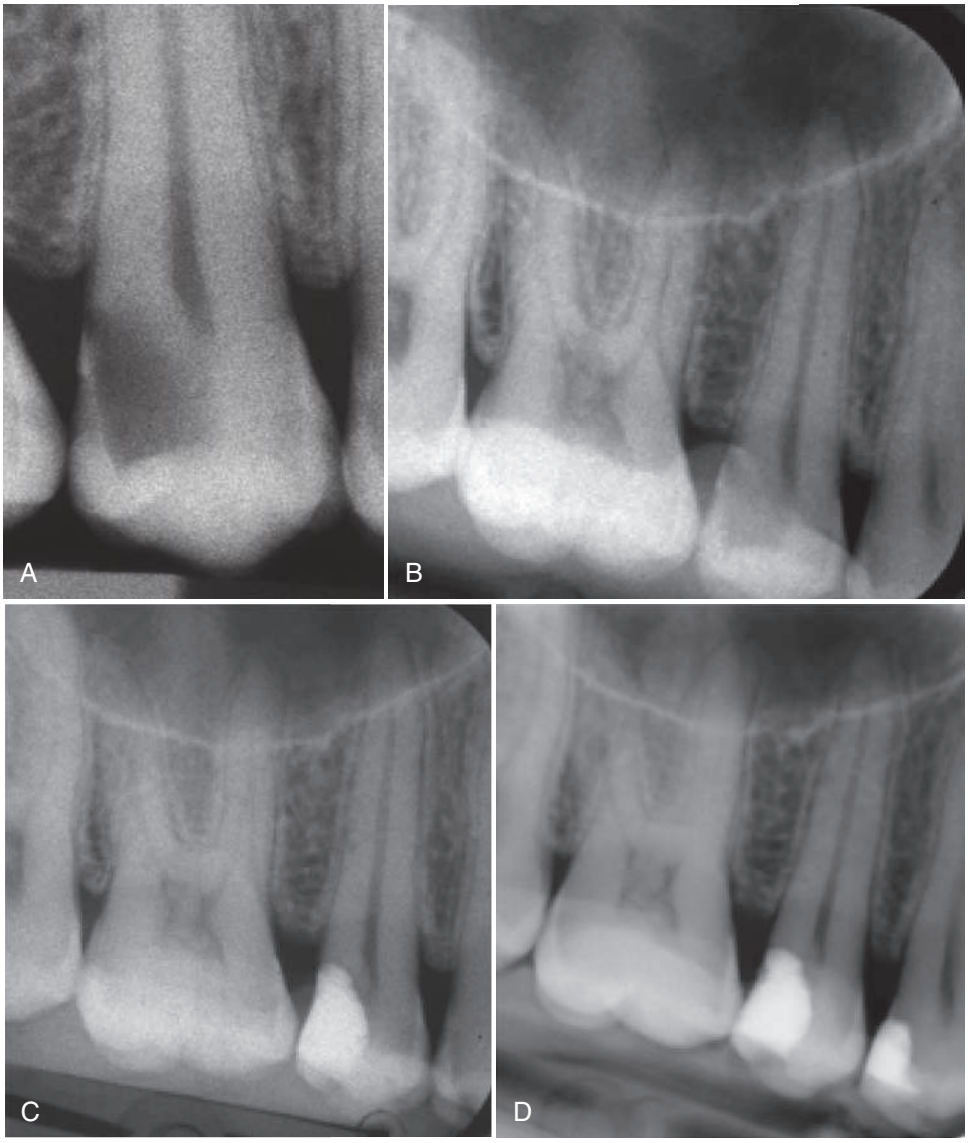


Fig. 24.3 A 24-year-old male patient presented with a deep distal carious lesion extending to the inner pulpal quarter in a maxillary right second premolar. The tooth was symptomatic but responded normally to sensibility testing. **A,** Magnified bitewing radiograph showing extension of advanced caries close to coronal pulp. **B,** Preoperative radiograph with period indentical device for precise comparison. **C,** Eighteen-month radiographic review after indirect pulp capping completed with calcium hydroxide. **D,** Five-year radiographic control. Pulp sensibility test response was positive with a well-defined lamina dura demonstrated on the periapical radiograph. (Reprinted with permission, © Lars Bjorndal 2019.)

exposures, the choice of Dycal (Dentsply, Konstanz, Germany) as capping material, and the use of long-term provisional restorations.[181] However, in the third study, where no pulp exposures were detected after complete caries removal, indirect pulp capping resulted in a 100% success rate at 12 months. The results indicate that CSCs may be preferable in indirect pulp capping procedures and that hard setting CH cements do not provide long- or short-term pulpal protection.[181]

PARTIAL PULPOTOMY

Partial pulpotomy (shallow or Cvek pulpotomy) is defined as the removal of a small portion of the vital coronal pulp as a means of preserving the remaining coronal and radicular pulp tissues.[14] During clinical treatment, the decision to remove a small or large portion of the coronal pulp is based on visual inspection of the tissue after pulp exposure during either caries excavation or exposure as a result of trauma (partial or shallow pulpotomy).[145] The incorporation of the dental operating microscope (DOM) is highly recommended. The coronal pulp is reduced from the exposed site by approximately 2 to 3 mm in order to remove necrotic or potentially inflamed and irreversibly damaged tissue, keeping the remaining pulp vital.[49] Partial pulpotomy is preferably performed with a high-speed handpiece and a small round diamond bur with copious water spray or continuous rinsing with physiological saline solution.[135,140,165]

After partial tissue removal, if bleeding cannot be controlled after 5 to 10 minutes of direct exposure to NaOCl, complete removal of the coronal pulp to the pulp floor is the preferred option.[211] NaOCl is a valuable diagnostic as well as clinical tool to differentiate irreversible from reversible

pulpitis in the coronal pulp in order to help determine whether to proceed with partial pulpotomy, complete pulpotomy, or pulpectomy. This decision can be of paramount importance in young permanent teeth with open apices, in which removal of tissue contaminated by microorganisms can reverse symptoms and stabilize inflamed tissue.[89,130,390]

A recent study that examined MTA complete pulptomies suggested that a hemorrhage "control time" parameter is not an accurate indicator of pulpal inflammatory conditions, contrary to other investigations.[242] However, the study did not benefit from recommended protocols including verified necrotic tissue removal using the DOM. Investigations have shown that the inflammatory response of pulp tissue challenged by bacterial by-products can progress several millimeters into the coronal pulp or advance further, thus affecting the radicular pulp tissue distant from the injury site.[320] The removal of 2 to 3 mm of exposed coronal tissue or radicular tissue to access the deeper, healthy tissue, in cases of trauma, carious exposures or complete pulptomies can ensure pulp survival (Fig. 24.4).[101]



Fig. 24.4 An 7-year-old male patient referred after trauma displaying a horizontal coronal fracture of his maxillary right central incisor #8 with a pulp exposure covered with a glass ionomer cements by the general dentist and a retained deciduous left central incisor *(F)*. **A,** Clinical photograph showing horizontally fractured right maxillary incisor and retained deciduous left central incisor. **B,** Periapical radiograph reveals open apex #8 and unerupted maxillary left permanent incisor #9. In addition, a mesiodens is visible. **C,** Photograph after shallow Cvek coronal pulpotomy. **D,** View of access cavity after sodium hypochlorite hemostasis and placement of 3 to 4 mm Biodentine (Septodont, Saint-Maur-des-Fossés, France) plug. **E,** Radiograph after pulpotomy and temporary restoration placement. **F,** Clinical photograph 3 days after verified calcium silicate cements setting and reattachment of the fractured coronal segment after bonding. **G,** Radiograph of #8 with bonded crown segment reattachment showing immature open apex.

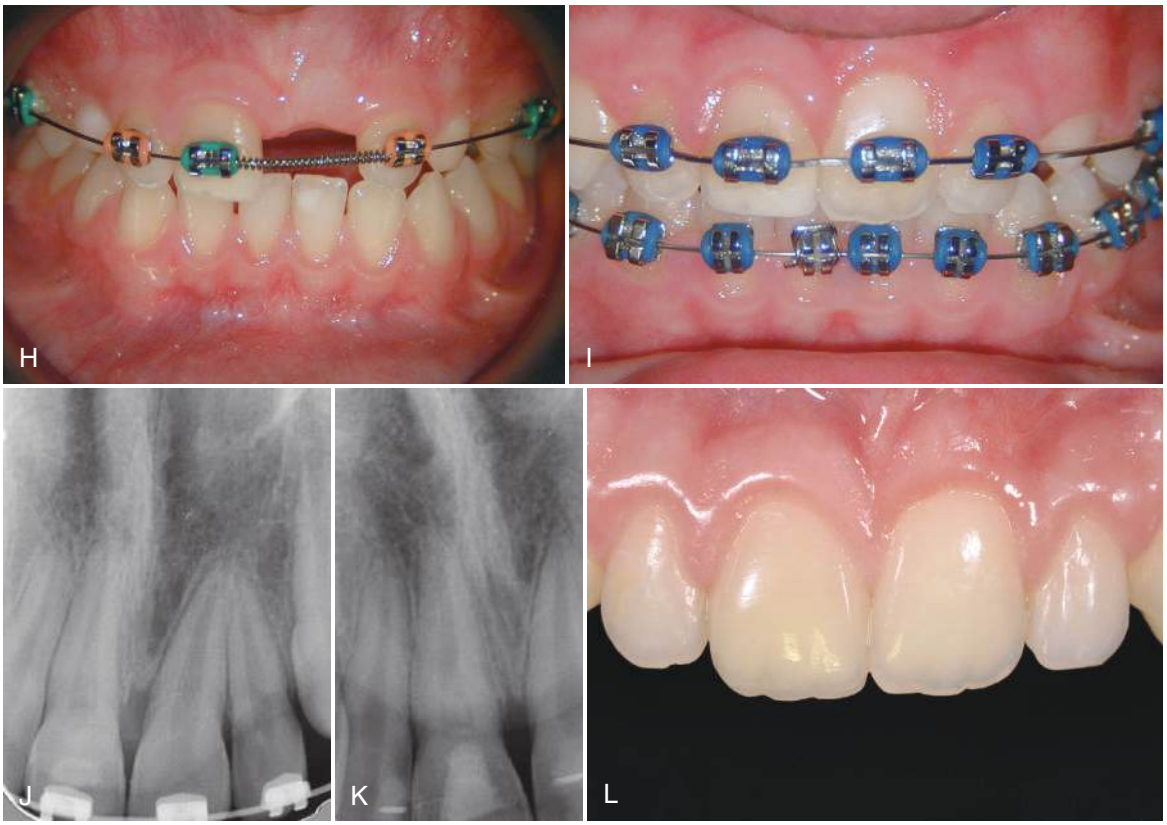


Fig. 24.4, cont'd **H,** Clinical photograph of full mouth banding after extraction of *(F)* and the mesiodens to permit normal eruption of tooth #9. **I,** Four-year clinical photograph with full orthodontic banding and complete eruption of #9 into the occlusal plane. Tooth #8 responded normally to cold sensibility testing at 1-, 2-, and 3-year recalls. **J.** Four-year periapical radiograph showing root-end closure of tooth #8 with normal periapical appearance. **K,** Five-year radiographic review of tooth #8 demonstrating complete radicular maturation. The incisor responded positive to cold testing. **L,** Clinical photograph at 5 years after finishing of a resin composite veneer created with a layering technique on tooth #8. (Courtesy Dr. Marga H. Ree with restoration by Dr. Caroline Werkhoven, Amsterdam, Netherlands.)

The amputation site is covered with a NaOCl soaked cotton pellet until hemorrhage ceases. If the formation of a blood coagulum is prevented, the same pulpal repair mechanisms associated with DPC are expected.[102,191] If the remaining pulp is healthy, bleeding can be stopped within 5 to 10 minutes. If hemostasis does not succeed within this time, it indicates that the pulp is irreversibly inflamed, and a complete pulpotomy (the removal of the entire coronal pulp) may be considered as a last resort to maintain vitality.[225] A CH suspension or CSC is applied to the newly exposed pulp surface and covered with a cement (for the CH suspension) or a thin layer of flowable composite, compomer, or RMGI (for the CSC).[92] A current study examining MTA versus CH pulpotomy has shown superior performance by MTA in mature permanent teeth compared with a hard-setting CH (Dycal). Since the capping material is used in larger quantities in the partial pulpotomy than in the DPC, the risk of tooth staining is considered higher when using a bismuth oxide containing CSC.[225] Partial pulpotomy and DPC are similar procedures, differing only in the amount of vital pulp tissue remaining after treatment. Partial pulpotomy is the preferred option in elective treatment procedures for teeth diagnosed with anatomic anomalies, such as dens evaginatus and invaginatus.

COMPLETE PULPOTOMY

Complete pulpotomy, or complete *pulp amputation,* is a more intrusive procedure defined by the AAE as "the removal of the coronal portion of the vital pulp as a means of preserving the vitality of the remaining radicular portion: may be performed as emergency procedure for temporary relief of symptoms or therapeutic measure, as in the instance of a Cvek pulpotomy."[14] The entire coronal pulp is removed and the vital pulp tissue capped at the entrance level of the root canal orifice. After successful hemostasis, it is treated analogously to DPC (Fig. 24.5).[225]

The assessment, selection, and amount of tissue removal are dependent on observer experience and completed on a case-by-case basis using magnification. Generally, after necrotic coronal tissue is completely removed, the inflamed pulp can be partially or completely amputated to the pulp floor or cervical area (pulpotomy) in the case of molars and some premolars.[144,147] The remaining tissue requires hemostasis before the placement of a CSC. The AAPD guidelines state, "A pulpotomy is performed in a tooth with extensive caries but without evidence of radicular pathology when caries removal results in a carious or mechanical pulp exposure."[13] However, the guideline requiring an absence of apical pathosis before pulp exposure for pulpotomy procedures may be outdated and require modification. Growing evidence from contemporary

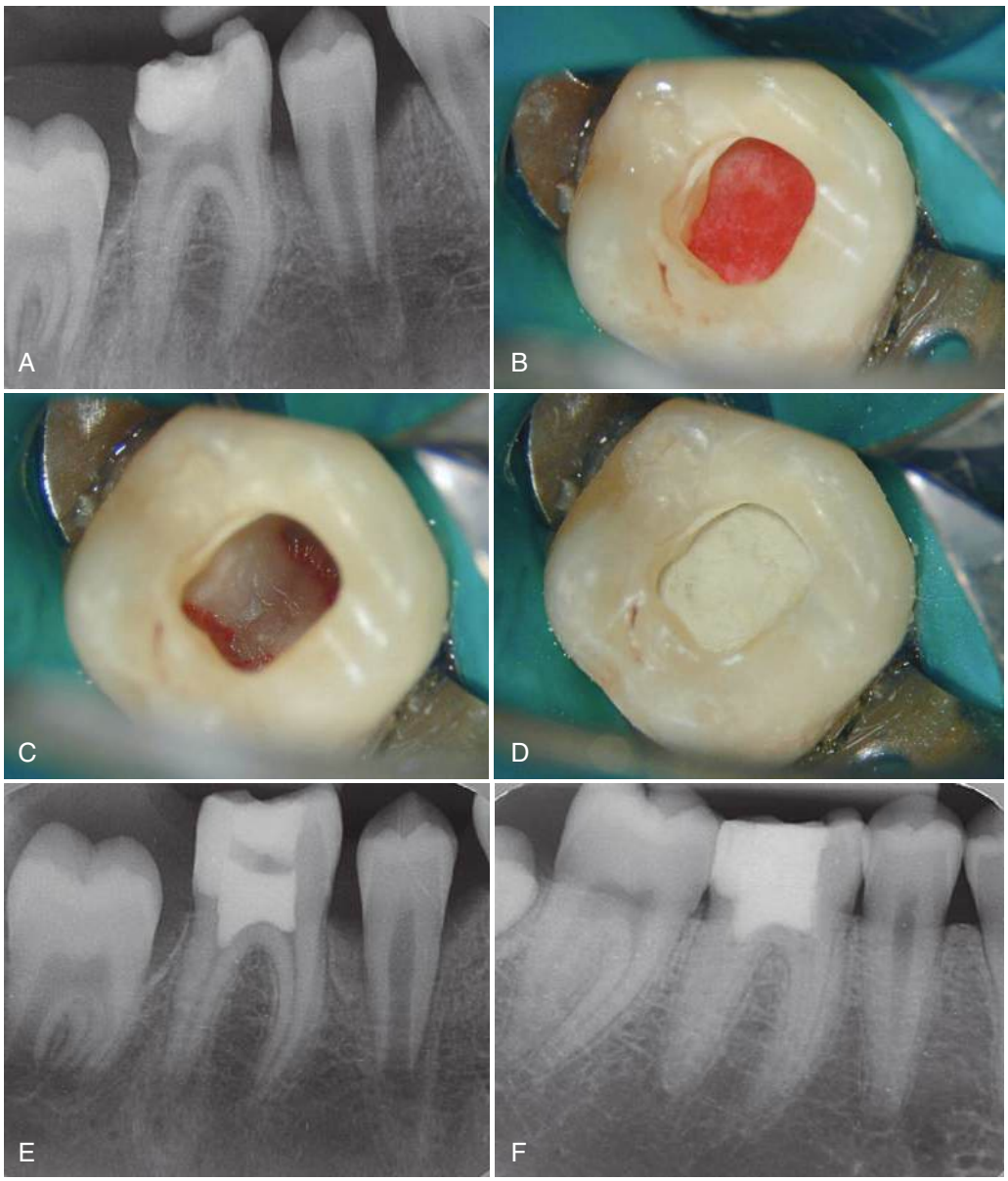


Fig. 24.5 A 10-year-old female was referred with a mandibular right first molar pulp exposure generated during caries excavation and temporized by the general dentist. **A,** Radiograph showing open apices with no evident apical pathosis. The tooth was diagnosed with reversible pulpitis. The patient received a new resin composite buildup to provide better isolation before a complete pulpotomy was performed. **B,** Clinical photograph of access opening through the buildup showing a sterile cotton pellet on the pulpal floor saturated with 5% sodium hypochlorite (NaOCl) for hemostasis. **C,** Photograph of radicular pulp in orifices after complete pulpotomy and 5% NaOCl hemostasis **D,** View through access opening after 5 mm thickness white mineral trioxide aggregate (MTA) (ProRoot MTA, Tulsa/Dentsply, Tulsa, OK) placement onto radicular pulp tissue and pulpal floor. **E,** Postoperative radiograph showing white MTA with wet cotton pellet and provisional restoration. One month later, MTA setting was verified and a permanent composite restoration provided. **F,** Radiographic review at 6 years shows apical maturation and thickening of the root walls. The patient remained asymptomatic and the tooth responded positively to electric pulp testing. (Reprinted courtesy Dr. Marga H. Ree, Purmerend, Netherlands, with permission.)

studies shows resolution of radicular lesions after pulpotomy treatments using CSCs in adult teeth diagnosed with irreversible pulpitis.[20,22,23,25,242,361–363]

Complete pulpotomy has traditionally been recommended for primary and immature permanent teeth for which short-term outcomes are generally favorable. However, the treatment of mature permanent teeth can be traced to Roman times around 200 BC.[415] Since the introduction of CSCs, there has been a renewed interest in pulpotomy as an alternative option to conventional orthograde root canal treatment, particularly in mature adult permanent teeth. Ongoing clinical investigations strongly support the use of CSCs for pulpotomy procedures in these teeth and show acceptable outcomes in pain reduction with continued normal function.[21–23,25,211,242,363,364,386]

Materials for Vital Pulp Therapy

For successful pulp capping outcomes, the removal of noxious stimuli, control of any infection, and the biocompatibility of the capping material are important prerequisites.[55,75] It has

been shown that rat pulp, in the absence of microorganisms, has the regenerative power to seal an exposed pulp with mineralized tissue even without medication or the use of restorative materials.[210,306] The presence or absence of microorganisms is the determining factor in the healing of pulp tissue.[210] The tissue reactions that follow capping (for instance, collagen synthesis and secretion) are generally similar to connective tissue injuries.[334,353] While excavating a deep caries lesion or during trauma, dentin or pulp tissue areas can be directly exposed, which are particularly permeable due to the structure of dentin. Closer to the pulp chamber, dentinal tubules occur more frequently per square millimeter and have larger diameters than further from the pulp.[336]

Since microorganisms, bacterial toxins (lipopolysaccharides), and many restorative materials can damage the exposed pulp tissue or diffuse via the dentinal tubules, causing irritation, both the pulp and the surrounding dentin should therefore be covered with the indicated capping material. Alternatively, pulp capping materials must create an artificial barrier between the vital pulp and the oral cavity to prevent the ingress of microorganisms. Moreover, the capping material should have antimicrobial properties without being toxic to the pulp.[108] In addition to the disinfection and sealing of dentin, another requirement of the capping material is hard tissue regeneration by pulp cell induction and thus preservation of pulpal vitality.[128,273,334] Mineralized tissue regeneration can occur in healthy pulps beneath a variety of capping materials.[15] The healing of pulp wounds is therefore not associated with a special drug or wound dressing.[44] However, medications to treat an exposed pulp should promote the innate capacity of pulpal cells to form hard tissue.[334]

AQUEOUS CALCIUM HYDROXIDE SUSPENSIONS

CH has long been considered the universal standard material for vital pulp therapy. Although the material demonstrates many advantageous properties, long-term study outcomes in vital pulp therapy have been inconsistent.[30,40,42,196] Desirable characteristics of CH include an initial high alkaline pH, which is responsible for stimulating fibroblasts and enzyme systems. It neutralizes the low pH of acids, shows antibacterial properties, and promotes pulp tissue defense mechanisms and repair. The drawbacks of CH include weak marginal adaptation to dentin, degradation and dissolution over time, and resorption in primary teeth. Histologically, CH demonstrates cytotoxicity in cell cultures and has been shown to induce pulp cell apoptosis.[15,159,335]

Tunnel defects have also been demonstrated in hard tissue barriers generated over pulp wounds associated with both CH and CSCs.[10,15,38,98,99,159,162,275,334] However, the primary difference between the two agents is that CH products are absorbable over time and dimensionally unstable. The slow disintegration of the CH after mineralized tissue formation can allow microleakage, permitting a slow ingress of microorganisms through defects. This can induce subsequent pulpal degeneration and lead to potential dystrophic calcification and pulp necrosis. Over extended periods, this problematic outcome can complicate any necessary nonsurgical root canal treatment at a later date.[274]

Clinical retrospective investigations have shown variable success rates over 2- to 10-year recall periods for direct CH pulp capping in humans.[30,40,42,196] Two current studies have examined the efficacy of CH as a DPC agent. One examined the survival of 248 pulp-capped teeth that were diagnosed either as having normal pulps or exhibiting spontaneous pain; the researchers found an overall survival rate of 76.3% with an average recall period of 6.1 years.[110] Outcomes were less favorable for teeth showing spontaneous pain, in older compared to younger patients, and in teeth restored with glass ionomer cements (GIC). The probability of pulps becoming nonvital after CH pulp capping was greater within the first 5 years of treatment.

The second study observed 1075 teeth directly pulp capped with a CH-based agent; these teeth had either healthy pulps or showed signs of reversible pulpitis.[288] Inclusion criteria limited pulp chamber roof exposures to no larger than 2 mm in diameter. Successful outcomes were 80.1% after 1 year and 68% after 5 years; this diminished to 58.7% after a 9-year observation period.[395] The results demonstrate increasing failure rates over time, attributable to absorption of the material under permanent restorations proximal to hard tissue bridges with tunnel defects. Another investigation has confirmed decreasing success rates with CH pulp capping with extended recall periods.[258] CH clearly has many favorable characteristics, but it can no longer be considered the preferred agent in vital pulp therapy (Fig. 24.6).[134,193,258,260,286]

HARD SETTING CALCIUM HYDROXIDE CEMENTS

In contrast to aqueous CH suspensions, other CH combinations such as cements (calcium salicylate ester cements) as well as liners or putties are less suitable for pulp capping due to a lower release of hydroxyl ions. In these preparations, the resulting pH is lower and the antimicrobial effect is significantly weaker.[139] In addition, hardening calcium salicylate ester cements (e.g., Dycal; Life, KerrHawe, Bioggio, Switzerland) are characterized by long-term disintegration.[38] They also fail to provide permanent support for the final restoration.[99] New mineralized tissue formation below CH-salicylate ester cements can be slow to generate and be less uniform in shape. Hard tissue regeneration may therefore be weaker using these CH preparations. In addition, inflammation is more common than with CH suspensions.[334] Some additives that harden the CH preparations may even have a toxic effect on the pulp.[99,334] For direct capping or pulpotomy, hard setting CH cements based on calcium salicylate esters cannot be recommended.

LIGHT-CURING LINERS AND CEMENTS

To compensate for the disadvantages of proven CH preparations, light-curing liners and cements with CH additives were developed (e.g., Ultrablend Plus, Ultradent, South Jordan; Calcimol LC, VOCO, Cuxhaven, Germany). These materials do not have sufficient mechanical strength despite their composite content, and specific CH effects are minimal.[359] Moreover, the pH of these products is significantly lower compared to other CH products.[359] Nonetheless, antibacterial effects can be detected in vitro, although this has not been confirmed by other studies.[309,402] The cytotoxicity of these products, which is due to the composite component, has been clearly demonstrated.[184,309,310] Therefore according to the

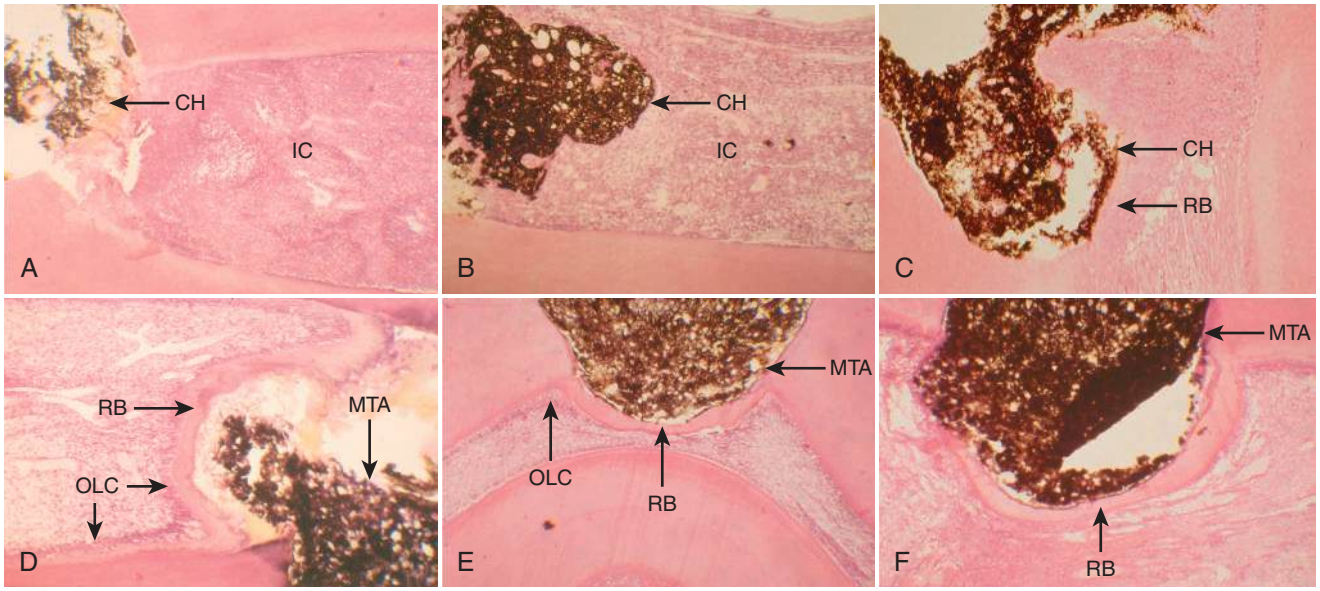


Fig. 24.6 Reparative bridge formation compared in dog pulps using mineral trioxide aggregate *(MTA)* and calcium hydroxide *(CH)*. **A,** Two-week pulp response to CH showing inflammatory cells *(IC)*. **B,** Tissue specimen showing lack of bridge formation and disorganized tissue proximal to CH. **C,** An 8-week specimen with partial reparative bridge *(RB)* formation subjacent to CH. **D,** Two-week pulpal response to MTA showing notable barrier formation and layer of organized odontoblast-like cells *(OLC)*. **E,** Pulp tissue section demonstrating complete calcific bridge formation proximal to MTA at 4 weeks. **F,** Sample section pulp capped with MTA at 8 weeks showing organized hard tissue formation with no inflammatory cell infiltrate. (Loma Linda University, Loma Linda, CA.)

manufacturer's instructions, the direct contact of Calcimol LC to pulp cells should be avoided.

Another light-curing product is TheraCal LC (Bisco, Schaumburg, IL). The spectrum of indications according to the manufacturer corresponds to that of MTA and other CSCs. Unlike all other CSC products, where a cement powder must be mixed with a liquid (water), TheraCal LC consists of 45% Portland cement CEM III and 45% composite resin and is therefore photopolymerizable, significantly simplifying its clinical use.[154,155] However, it is known that all the monomers in TheraCal LC are cytotoxic in vitro.[180,202,207] After 1 day of direct contact with TheraCal LC, the fibroblast cell turnover rate decreased by 31.5% and after 1 week to 45.9%.[184] After 72 hours of contact with TheraCal LC there was a marked decrease in the percentage survival of the cells.[310] For tissue healing (hard tissue formation after pulp capping), it is crucial that there is no decrease but an increase in the cell turnover rate. In contrast to TheraCal LC, other calcium silicate products and CH have been shown to induce pulpal cell proliferation.[112,113]

Similar to other light-curing liners using CH, TheraCal LC exhibits a significant cell-damaging effect when in direct contact with human dental pulp stem cells.[58,184,309,310] It is also unclear how much Portland cement can be released from the resin matrix when the material is polymerized. After curing, TheraCal LC shows a heterogeneous structure with a high proportion of large, unhydrogenated particles, since the resin additives do not provide enough moisture to allow hydration. The hydration of TheraCal LC (the reaction of cement with water) is incomplete because of limited moisture diffusion within the material. Thus, no CH and only a small amount of calcium ions are released.[70] Light-curing liners and cements with added CH or MTA are collectively cytotoxic.[309,310] It is currently inadvisable to use photopolymerizable CH or calcium silicate-containing materials for indirect or DPC.

COMPOSITE RESINS, DENTIN ADHESIVES, AND RESIN-MODIFIED GLASS IONOMER CEMENTS

Adhesive systems were introduced in the early 1980s as potential agents for DPC of cariously and mechanically exposed pulps.[200,253,404] Dentin adhesives, hydrophilic resins, and RMGI cements initially showed favorable outcomes in preliminary pulp capping investigations with nonhuman primates, based on standards set by the International Organization for Standardization.[96,367–369] However, the transitional use of these materials in human subjects did not demonstrate the corresponding biocompatibility or consistent reparative bridge formation.[5,119,159,183,197,271] Investigations that have examined responses to resin-based materials in human teeth have demonstrated unfavorable histologic reactions when the material is placed directly or in close proximity to pulp tissue.[5,124,172,272,315] Histological material typically demonstrates inflammatory cell infiltrates consistent with pulp cell cytotoxicity, subclinical adhesive failures at the pulp interface, and a profound absence of biocompatibility.[5,119,272,296,315,349]

The application of dentin adhesive to a thin layer of dentin (0.5 mm) leads to an expansion of the blood vessels and chronic inflammation of the pulp tissue.[183] Nonpolymerized or partially polymerized dentin adhesives cause apoptosis in various pulp cells, such as macrophages and undifferentiated pulp cells.[246] Besides monomers from dentin adhesives, identical cytotoxic components can also be released from composites, which then penetrate through dentinal tubules to pulp tissue causing damage.[59] Also, deep cavities (without pulp exposure) are problematic because the residual dentin thickness over the pulp can be thin, and tubular fluid can

prevent complete polymerization. In addition, complete polymerization can be prevented by an excessively thick composite layer that is insufficiently penetrated by curing lights.[272] Both factors lead to an increased proportion of residual-free monomers, which are then released from the composite resin and diffuse into the pulp tissue. In clinical studies the use of dentin adhesives in deep cavities results in relatively frequent pulp necrosis.[337]

Unpolymerized, polymerized, and aged dentin adhesives release cytotoxic components that are responsible for inferior performance in human and animal experiments.[5,94,112] In addition, monomers from restorative materials (dentin adhesives/composites) may interact with the immune system of the pulp, weakening its defense against invading microorganisms.[44,268] Direct contact with dentin adhesives or composite resins is thought to reduce pulpal defense, either due to cytotoxicity or specific changes in the immune response.[207,267,268] Adhesive components thus inhibit the proliferation of immunocompetent cells and cause chemical immunosuppression, which promotes the development of pathological changes in the pulp.[44,207] Dentin adhesives and composites are essentially not biocompatible and should not be used for indirect or DPC.[9,94] However, hydrophilic resins and RMGI cements provide excellent seals when they are combined with light-cured composites as permanent restorations and placed directly over pulp capping materials such as MTA and other CSCs.[27,131,288]

MINERAL TRIOXIDE AGGREGATE

MTA was introduced as a pulp capping agent by Torabinejad and associates in the mid-1990s.[141] Most preliminary experimental and current clinical data in vital pulp therapy are based on the proprietary material ProRoot MTA (Tulsa/Dentsply, Tulsa, OK). The cement consists of a hydraulic calcium silicate powder containing oxide compounds, including calcium oxide, ferric oxide, silicon oxide, sodium and potassium oxides, magnesium oxide, and aluminum oxide.[71] The material exhibits favorable physicochemical characteristics that stimulate reparative dentinogenesis by recruitment and activation of hard tissue–forming cells, contributing to matrix formation and mineralization.[298] Soluble cytokines and growth factors that mediate wound repair of the dentin-pulp complex are nested in the extracellular matrix. MTA stimulates reparative hard tissue formation by sequestering these growth factors and cytokines embedded in the surrounding dentin.[222,223,375,385]

CH and calcium silicate hydrate, the principal by-products formed during hydration of mixed MTA, contribute to a sustained alkaline pH.[68,143] The setting properties of the hydroscopic silicate cements are not affected by the presence of tissue fluids or blood.[377] They are therefore also referred to as "hydraulic" cements, as they are hard setting both in contact with air and water.[47] During the setting process, the gradual release of calcium ions encourages reparative barrier formation by promoting signaling molecules, such as VEGF, macrophage colony-stimulating factor, TGF-β, and interleukins IL-β and IL-1α.[251,300]

MTA demonstrates superior marginal adaptation to dentin compared to CH-based agents and forms a dense bond to dentin. Mineral components from the cement penetrate into the tubules, producing adhesion to dentin comparable to GICs.[29,216] MTA also forms an adherent interfacial layer during mineral nucleation at the dentin surface that appears similar in composition to hydroxyapatite when examined with x-ray diffraction, energy-dispersive x-ray analysis, and scanning electron microscopy (SEM).[298,329,379] Similar to CH, MTA induces an inflammatory cascade that results from calcium ion release and the creation of an alkaline environment, producing tissue necrosis. Both MTA and CH have been shown to stimulate and increase the Höhl cell mitosis index in rodents.[114,115]

MTA activates the migration of progenitor cells (fibroblasts) from the central pulp to the injury site and promotes their proliferation and differentiation into odontoblast-like cells without inducing pulp cell apoptosis.[298] MTA also stimulates in vitro the production of messenger RNA and increases protein expression of the mineralized matrix genes and cellular markers crucial for mineralization after matrix formation.[298] In a human study, odontoblast-like cells and the formation of a dentin or dentin-like tissue could not been proven after DPC with MTA (RetroMTA; BioMTA, Daejeon, Korea). Hence the new calcified hard tissue was not "regular dentin" and did not seem to be the product of genuine odontoblast differentiation. This observation is identical to results after DPC with CH. Hard tissue formation after DPC appears to be a dystrophic intrapulpal mineralization in response to the capping procedure.[111]

Gray MTA has been shown to enhance cell proliferation and survival of cultured human dental pulp stromal cells.[300] The biocompatibility of set MTA upregulates the expression of transcription factors, angiogenic factors, and gene products, such as dentin sialoprotein, osteocalcin, and alkaline phosphatase.[299] Odontoblast signaling proteins are essential in the differentiation of progenitor cells into cells responsible for repair and hard tissue deposition.[299,300] After MTA pulp capping, both sialoprotein and osteopontin have been observed in the mineralized hard tissue matrix at the exposure site during the process of reparative hard tissue formation.[227]

Dental pulp cells differentiate into the odontoblastic cell line in the presence of the signaling molecules, such as TGF-β, heme oxygenase-1 enzyme, and bone morphogenetic proteins BMP-2, BMP-4, and BMP-7.[171] MTA most likely upregulates fibroblast secretion of BMP-2 and TGF-β1 and therefore stimulates and promotes mineralization and hard tissue regeneration.[117,170,171,178,340,365,406] MTA induces a time-dependent environment that is proinflammatory and promotes wound regeneration through upregulation of cytokines.[318] Immunohistochemical analyses show that cytokines, including myeloperoxidase, inducible nitric oxide synthase, VEGF, nuclear factor-kappa B, activating protein-1, and cyclooxygenase-2, show increased expression in the presence of MTA. Cytokine upregulation is responsible for inducing biomineralization by producing apatite-like clusters on collagen fibrils at the MTA-dentin interface. MTA does not affect the generation of reactive oxygen species, thereby positively influencing cell survival. MTA also has been shown to improve the secretion of IL-1β, IL-6, and IL-8.[67,81,91] However, an inhibitory effect on dental pulp cells has been demonstrated in the presence of MTA, which may be attributed to the release of aluminum ions.[267] Overall, the

data indicate that MTA promotes a biocompatible, noncytotoxic, antibacterial environment and surface morphology that is favorable for reparative calcific bridge formation. MTA stimulates the release of the dentin matrix components necessary for hard tissue repair and regeneration in mechanically exposed healthy and partially inflamed pulps (Figs. 24.7–24.9).[6,141,170,266,286,317,374]

One disadvantage of MTA is that it can lead to discoloration of the hard tooth tissue. This may be problematic, particularly in anterior teeth during trauma management.[279] It is due to heavy metals contained in MTA, such as bismuth oxide used for radiopacity or iron.[45,123,347] The discoloration is mainly induced by the oxidation of these metals after contact with NaOCl or the uptake of blood components.[69,234,347] Other CSCs contain less or minute levels of heavy metals and their potential to discolor teeth is minimal. CSCs containing zirconium oxide or tantalum oxide, which are used as radiopacifiers, are particularly color-stable.[279]

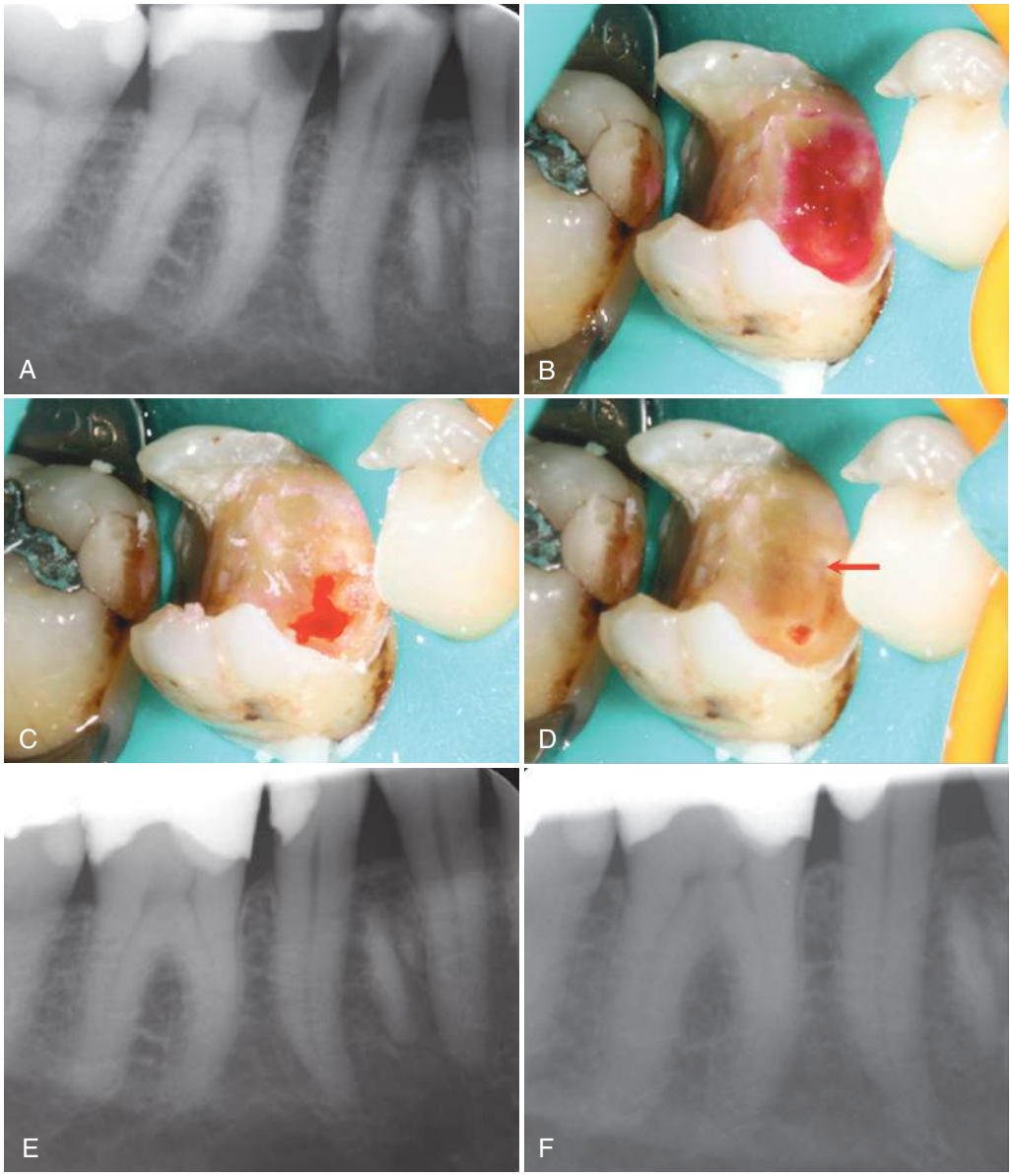


Fig. 24.7 A 50-year-old male patient with deep recurrent caries associated with a mandibular right first molar (#30). The patient had no history of pain and responded normally to cold testing. **A,** Periapical radiograph showing distal caries in the mandibular right second premolar (#29) and large carious lesion below mesial aspect of composite restoration in the molar. Note retained deciduous root tip mesial to tooth #29 and normal periapical appearance of #30. **B,** Pulpal hemorrhaging evident after complete caries removal with hand excavators. **C** and **D,** Clinical photographs after bleeding was controlled using cotton pellets soaked in buffered 0.5% sodium hypochlorite (NaOCl). Note presence of reparative dentin protecting mesiolingual pulp horn *(red arrow)*. A 2 mm–thick layer of white mineral trioxide aggregate (Dentsply, Tulsa Dental, Tulsa, OK) placed directly over the exposure and surrounding dentin and covered with a thin wet cotton pellet. The tooth was provisionalized for 1 week and restored permanently with composite resin. **E,** One-year radiographic follow-up; tooth responded normally to cold sensibility testing. **F,** Control radiograph at 3 years. Patient was asymptomatic and exhibited normal pulp testing response. (Courtesy Dr. Rita Kundzina, Tromsø, Norway.)

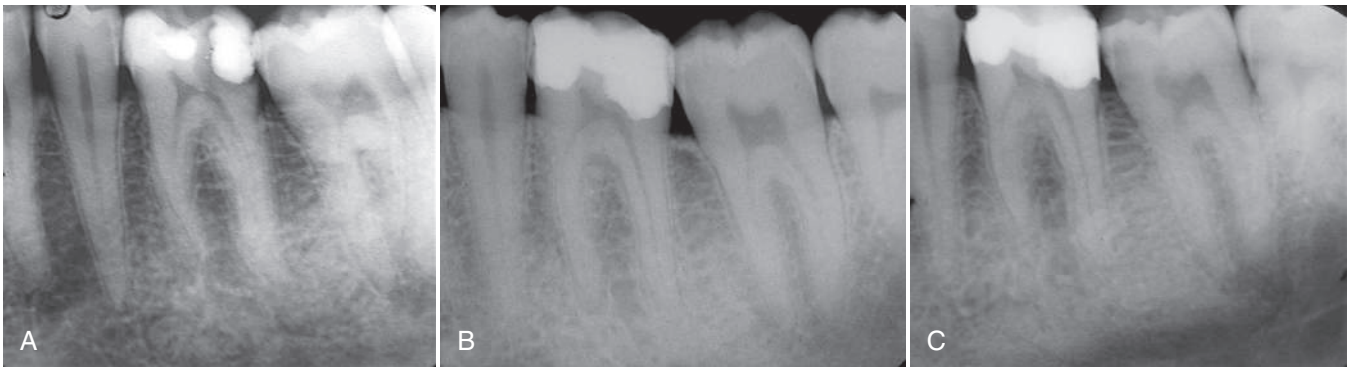


Fig. 24.8 Symptomatic mandibular left first molar in a 32-year-old male patient with temporary restoration placed over deep caries on the distal aspect and referred for root canal treatment. Sensibility testing elicited a normal pulpal response. **A,** Periapical film showing deep caries extending towards the distal pulp horn. **B,** Radiograph of direct mineral trioxide aggregate (Dentsply, Tulsa Dental, Tulsa, OK) pulp capping after 5.25% sodium hypochlorite hemostasis and placement of a three-surface bonded composite restoration. **C,** Nine-year 6-month radiographic control. The tooth was positive to cold testing and without symptoms. Note presence of mesial carious lesion in the second molar. (© Dr. George Bogen.)

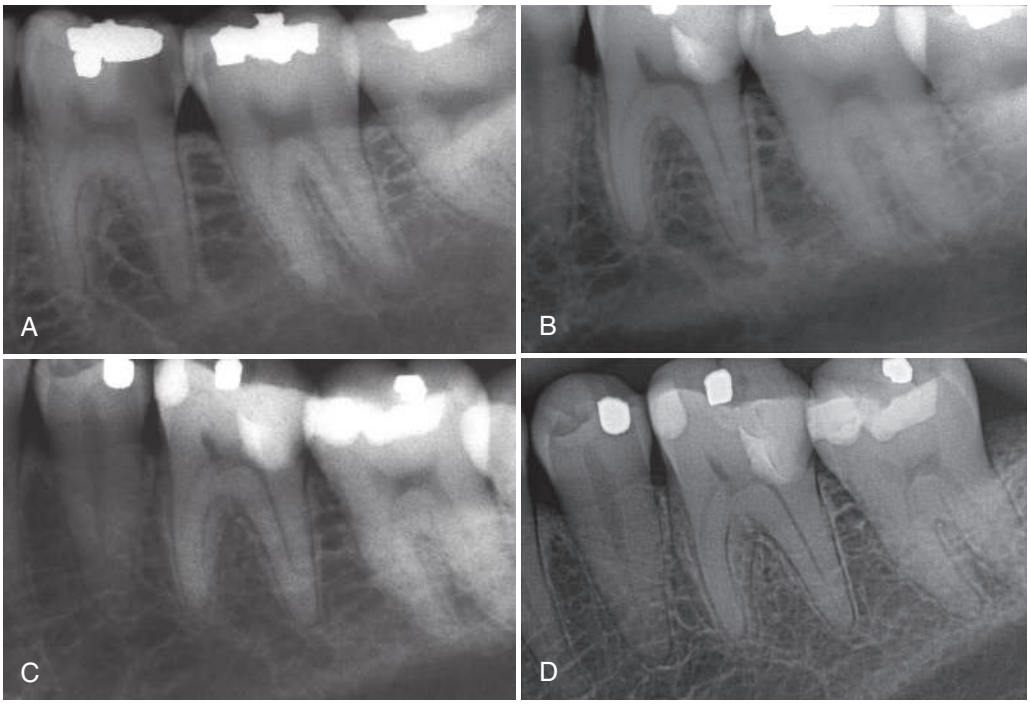


Fig. 24.9 **A,** Periapical radiograph of deep caries in a partially symptomatic mandibular left first molar in a 29-year-old female patient. **B,** Direct pulp capping completed with white mineral trioxide aggregate (MTA); the final bonded restoration was placed during the second visit after MTA curing. **C,** Radiographic recall at 7 years. Tooth was asymptomatic with normal response to CO_2 ice. **D,** Thirteen-year 10-month recall radiograph showing normal periapical appearance. The molar responded positive to cold testing and demonstrated normal mobility and periodontal probings. (© Dr. George Bogen.)

CALCIUM SILICATE CEMENTS

A variety of new CSCs have developed since the introduction of MTA.[116,153,232,302] Preliminary investigations have demonstrated physicochemical and bioinductive properties comparable to MTA, indicating promise for their application in vital pulp therapy.[18,57,127] Some of these tricalcium-based materials include BioAggregate (Innovative Bioceramix, Vancouver, BC, Canada), Biodentine (Septodont, Saint-Maur-des-Fossés, France), MTA-Angelus (Angelus, Londrina PR, Brazil), MTA Bio, and MTA Branco (Angelus).[160,237] Other formulations include EndocemMTA (Maruchi, Wonju-si, Gangwon-do, South Korea) and EndoSequence BC root repair material (Brasseler USA, Savannah, GA). Currently, there are more than 40 different CSC formulations from international manufacturers. Some other silicate cement products include EndoBinder, iRoot BP, MicroMega MTA, MTA Bio, MTA Plus, NeoMTA Plus, RetroMTA, Tech Biosealer Capping, TheraCal LC, CEM Cement (calcium-enriched mixture), Grey MTA Plus, MEDCEM MTA, EndoCem Zr, Channels MTA, DiaRoot Bioaggregate, and Harvard MTA.[302] Additional compounds are currently undergoing clinical investigations to establish their safety and efficacy.[219,278,302,378]

The main components of MTA and the new CSCs are tricalcium silicate and dicalcium silicate, the major components of

Portland cement. Hydraulic tricalcium silicates promote reparative barrier formation by the upregulation of transcription factors after gaining immediate strength on hydration. The cements also encourage hydroxyapatite crystal formation on the cement surface when in contact with calcium- and phosphate-containing fluids.[155,179] Moreover, the release of CH from CSCs during hydration has a positive effect on cell regeneration. Osteoblasts, cementoblasts, periodontal ligament cells, and pulp cells are deposited directly on the CSC surface, as the material is recognized as "non-foreign," which affirms the high biocompatibility of these cements.[378]

In addition to CH, silicon is released during CSC cement hardening. The exact function of silicon in the metabolic processes of hard tissue formation is unclear, but it is believed to play a role in the early stages of mineralization.[80] It is also known that silicon positively influences the rate of new hard tissue deposition when it is released from bioactive materials such as MTA.[304] In addition, silicon can remineralize demineralized dentin in vitro and this has also been demonstrated for CSCs.[28,304,327] It can be concluded that the release of silicon from calcium silicate-containing materials contributes to the induction of mineralized tissue formation.[179]

BioAggregate is a bioinductive tricalcium cement that can induce mineralization in osteoblast cells by increasing levels of osteocalcin, collagen type 1, and osteopontin gene expression.[412] Hydration of the cement results in the formation of calcium silicate hydrate and CH, showing high concentrations of silica and calcium phosphate.[166] This characteristic is consistent with materials used in vital pulp therapy to promote hard tissue formation. Investigations using x-ray diffraction show the material to have a composition similar to that of MTA, but BioAggregate contains tantalum rather than bismuth oxide for radiopacity.[303] It demonstrates remarkable biocompatibility compared to MTA, inducing cell differentiation in both human periodontal ligament and gingival fibroblasts.[361,405] Both fresh and set mixtures of MTA and BioAggregate have shown antimicrobial properties effective against *Enterococcus faecalis* in vitro when combined with equal amounts of human dentin powder.[414] The material also shows a greater resistance to dislodgement in an acidic environment compared to MTA, and higher fracture resistance when used as a filling material.[182,383]

Biodentine is a tricalcium silicate–based cement that also demonstrates exceptional bioactive properties.[2] Whereas MTA is more or less comparable to a refined Portland cement, Biodentine consists mainly of pure tricalcium silicate (about 80%) with calcium carbonate as filler (about 15%).[47,109] Zirconium oxide is added as radiopacifier (about 5%). In contrast to other CSCs and MTA, Biodentine does not contain dicalcium silicate or metal oxides except the radiopacifier.[72,341] The liquid consists of calcium chloride in an aqueous solution with an admixture of modified polycarboxylate. The powder is mixed in a triturator with the liquid in a capsule. Once mixed, Biodentine sets in about 15 minutes minimum, and hardens completely in 85 minutes.[217]

Biodentine does not induce genotoxic or cytotoxic effects when measured with the Ames mutagenicity test. It is considered a biocompatible dentin replacement material for use under restorative materials as a base.[232] During setting, Biodentine releases calcium ions forming CH that lead to an alkaline pH in the surrounding tissues, inhibiting the growth of microorganisms.[48,70] Furthermore, when Biodentine is applied onto pulp cells ex vivo, TGF-β1 is released.[231,250] In vitro Biodentine induces proliferation, migration, and differentiation of human dental pulp cells and induces cell proliferation in osteoblasts and periodontal ligament cells.[17,209,232,307]

These factors may explain hard tissue formation after DPC with Biodentine.[2] The material stimulates biomineralization and encourages hard tissue formation when used as a capping material.[343,413] Histological evaluation in humans revealed complete mineralized tissue formation 6 weeks after DPC with mild to absent pulp inflammatory reactions, comparable to MTA.[295,297] Biodentine forms needle-like crystals resembling apatite at the dentin interface.[120] Identically to MTA, Biodentine seals dentin by micromechanical retention with a shear bond strength comparable to GICs.[29,216] Different light-curing adhesive lining materials are able to bond to Biodentine and may prevent bacterial leakage.[2,333] After indirect capping with Biodentine, a reversible pulpitis can reproducibly heal.[182] Success rates of human DPC with Biodentine are between 82.6% and 86% after 1.5 to 2.3 years and thus in the range given for ProRoot MTA (Figs. 24.10 and 24.11).[181,193,206,226,244,248,260]

Another promising material for vital pulp therapy is MTA-Angelus, which has a basic formulation of 25% bismuth oxide and 75% Portland cement. The composition eliminates calcium sulfate, providing a short setting time of 15 minutes, preferable for pulp capping or pulpotomy procedures.[57] Variations in bismuth oxide and the presence of iron characterize the chemical composition of MTA-Angelus, and the crystalline structures formed on hydration are similar to gray and white ProRoot MTA.[355] These materials have been compared experimentally as pulp capping agents in intact, caries-free human teeth. Histomorphologic examination of the extracted teeth revealed similar inflammation and hard tissue formation responses.[4] MTA-Angelus also demonstrates antifungal properties and a lower compressive strength than ProRoot MTA.[41,212,219] Many CSC materials demonstrate acceptable antibacterial products; however, *E. faecalis* has shown resistance to some cements when measured with disc diffusion tests.[219]

EndoSequence BC root repair material shows low cytotoxicity, antibacterial activity against *Enterococcus faecalis*, and potential as a pulp capping material.[105,416] Another material, calcium-enriched mixture (CEM) cement, has also demonstrated excellent physical and biologic properties in vital pulp therapy investigations.[20–24,26,294] The advantages of these calcium silicate materials over the typically used CH products lie in the higher mechanical strength, lower solubility, and tighter sealing to dentine. Major disadvantages of CH are avoided when using CSCs: dissolution of the capping material as well as the mechanical instability and consequent lack of long-term protection against bacterial microleakage.[108] The new generation of CSCs appear promising when used as vital pulp therapeutic agents, and current investigations appear to support their future potential and expanded use in vital pulp therapy.

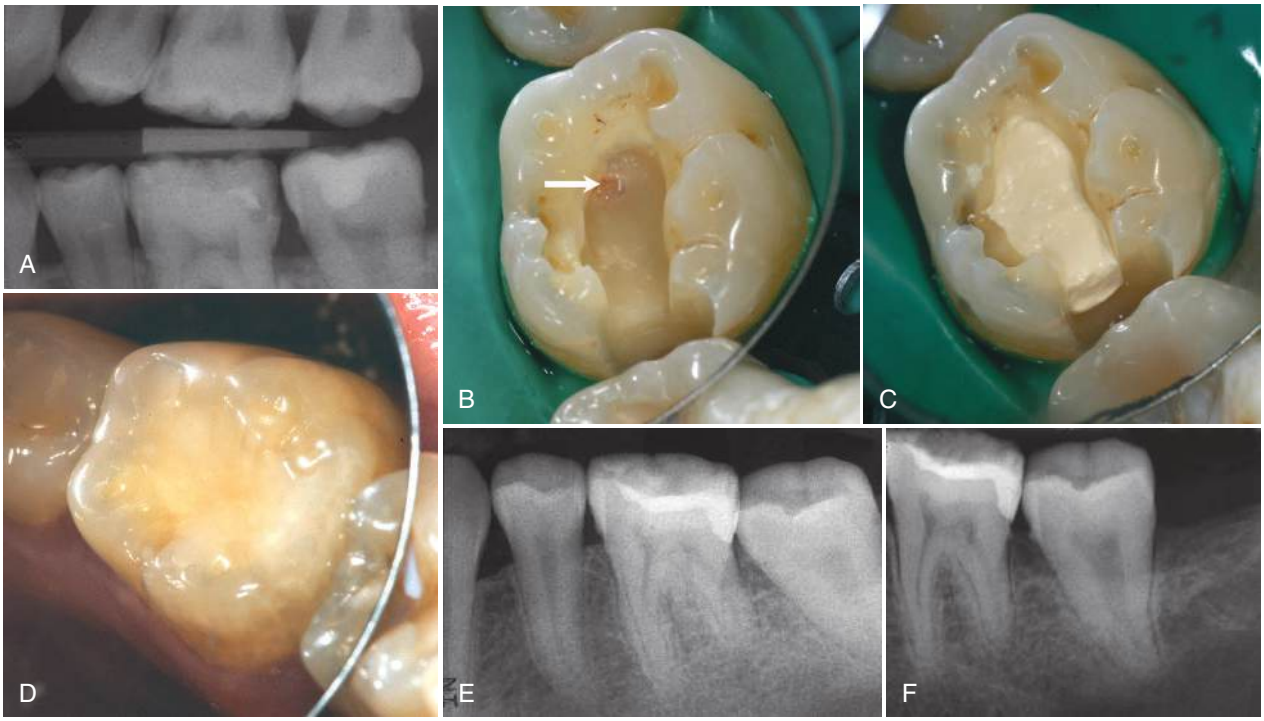


Fig. 24.10 A, Bitewing radiograph of quadrants 2 and 3 showing defective restoration in a mandibular left first molar (#19) of a 27-year-old female patient. **B,** Clinical photograph showing pulp exposure *(white arrow)* after cavity preparation and complete caries excavation. **C,** Pulp capping and restorative base completed under slight pressure using Biodentine (Septodont; St. Maur-des-Fossés, France). **D,** Clinical photograph of final two-surface composite restoration placed after a setting time of 15 minutes. The restoration was placed after acid etching and bonded with a two-bottle dentin adhesive system. **E,** One-year periapical radiograph showing a normal periapical region and Biodentine radiopacity similar to that of surrounding dentin. **F,** Three-year radiographic review shows no evident pathological changes apically. The tooth registered normal responses to sensibility testing at both recall periods. (Reprinted with permission, Dr. Till Dammaschke and © Quintessence Deutschland.)



Fig. 24.11 A, Bitewing radiograph of quadrants 1 and 4 of an 18-year-old patient displaying a distal carious lesion in the maxillary right second premolar. The patient did not present at the scheduled appointment for restorative care. **B,** The patient returned 4 years later for treatment. Clinical photograph showing iatrogenic pulp exposure at two sites during complete caries excavation. **C,** Photograph of direct capping and base placement with Biodentine (Septodont; St. Maur-des-Fossés, France) applied to the cavity with cement pluggers using light pressure. **D,** Clinical view of cavity restored with bonded composite resin using a two-bottle dentin adhesive. **E,** Radiographic control recorded 4 years after direct capping showing normal periapical tissues. Tooth was asymptomatic with normal response to sensibility testing. (Reprinted with permission, Dr. Till Dammaschke and © Septodont [St. Maur-des-Fossés, France].)

Calcium Silicate Cement Applications in Vital Pulp Therapy

DIRECT PULP CAPPING WITH CALCIUM SILICATE CEMENTS

Controlled prospective investigations on DPC in humans using MTA and other CSCs against cariously exposed pulps are limited but expanding. Collectively, most studies are inconsistent with regard to case selection, treatment strategies, and clinical protocols.[268] The spectrum of outcomes reflects the absence of standardized guidelines for caries removal, hemostatic agents, choice and placement of the capping material, and single- versus two-visit delivery sequences. With the advent of quicker setting CSC formulations, indications for two-visit pulp capping are declining in favor of one-visit procedures. Moreover, although the concept of DPC in a carious field is gaining universal acceptance using MTA and other CSCs, clinical recommendations for avoiding and treating pulp exposures in permanent teeth as an accepted management strategy persist, despite contrary evidence.[129,316] Current international guidelines are variable because the research and technological advances within the field are moving exponentially without a universal consensus.

Human studies using CSCs other than MTA are showing the mutual biocompatibility of new bioceramic materials, indicating that conjointly, the cements exhibit similar properties and cellular responses when exposed to vital pulp tissues. Comparable outcomes are evident when pulp capped teeth are examined for mineralized tissue barrier formation, lack of inflammatory cells, and continued vitality in DPC.[31,62,181,185,206,211,214,219,243,301] Analysis has shown that initial pulp capping treatment is also cost-effective compared to root canal treatment in younger patients with pulp exposures due to caries.[63]

Current and past investigations comparing newer CSC formulations have shown clinical and radiographic success in permanent teeth treated with carious exposures ranging from 83.4% to 100% at 1 to 7.4 years.[31,62,181,185,214,243,244,301] In a larger study using Biodentine as a DPC agent, 245 teeth recalled with an average recall period of 2.3 years showed continued vitality in 86% of the cases. In one study where Biodentine was compared against MTA in a split mouth model or randomly allocated in different patients, outcomes at recall periods of 18 to 36 months were comparable for both materials when examined in permanent mature teeth with carious exposures.[214] Impressive results have also been demonstrated in several other contemporary studies.[3,6,238,258] An investigation examined 30 immature permanent teeth exhibiting wide-open apices directly pulp capped with MTA and restored temporarily with intermediate restorative material (IRM).[138] Definitive permanent composite restorations were placed 2 weeks later, after confirmation of pulp vitality. In these cariously exposed immature permanent teeth, the success rate at a 2-year review was 93%.

One notable factor revealed in a recent study with a broad sample size was that treatment outcomes were negatively influenced when restored with a GIC compared to all other restorative materials.[181] The outcome is a likely indicator that GIC should not be considered as a definitive restoration in DPC cases.[181] The outcomes of these current studies strongly suggest that MTA and introduction of new CSC capping agents show capable biocompatibility and sealing properties and can make the outcome of DPC techniques in mature permanent teeth with carious exposures increasingly reliable.

Another observational investigation examined DPC of carious exposures of mature and immature permanent teeth using slow-set MTA in a two-visit sequence.[56] Forty-nine teeth were examined in patients aged 7 to 45 years over a 1- to 9-year period, with an average 3.94-year observation time. The study protocol used detector dye-aided caries excavation using the DOM, 5.25% to 6% NaOCl hemostasis, bulk MTA placement on pulp exposures, and surrounding dentin, coupled with bonded composite restorations placed at a second visit to compensate for the delayed setting properties of ProRoot MTA. Based on subjective symptomology, cold testing, and radiographic evaluation, 97.96% of teeth showed continued vitality. All patients with immature apices showed continued root formation and apical closure within a 1- to 6-year period (Fig. 24.12).

A related DPC investigation using a two-step protocol in permanent teeth using MTA with bonded restorations with slight variations in procedural execution registered similar outcomes. Sixty-four teeth recalled after 3.6 years showed an overall success rate of 91.3%. The success in patients older than 40 years of age was 80% and for those younger than 40 years was 100%.[248] In a corresponding study, 33 patients with DPC after carious exposures and provisionalized for 1 week (two-visit protocol) showed a 85% success rate at 3 years.[226] The same study also compared outcomes with patients that received direct CH capping and demonstrated a survival rate of 52% at 3 years. The majority of human studies that compare MTA/CSC direct capping in permanent teeth with CH suggest MTA/CSCs offer more predictable long-term benefits.[78,226,239] Moreover, larger scale pulp capping studies comparing MTA/CSCs with CH corroborate previous data that demonstrate superior results when MTA/CSCs are applied in a clinical setting (Figs. 24.13 and 24.14).[66,193,238,260]

The unique physicochemical properties of MTA and CSC also promote a superior environment for pulpal repair and bridge formation, compared to CH products.[193,238,260,286] MTA and CSC are hygroscopic cements that set in the presence of blood and serum, produce a gap-free interface with dentin, and generate a sustained alkaline pH; in addition, the surface morphology of the hardened cement allows for predictable bonding with current adhesion systems. Growth factors necessary for hard tissue formation are activated by CSCs through the gradual release of calcium ions during curing.[222] The small particle size and alkaline pH contribute to the entombment of remaining cariogenic bacteria at the dentin-MTA and CSC interface, impeding bacterial ingression and caries progression and discouraging continued pulpal injury (Fig. 24.15).[288,409,414]

PULPOTOMY WITH CALCIUM SILICATE CEMENTS

It has been recommended that pulpectomy be avoided in immature permanent teeth with vital canal tissue so as to protect the remaining radicular pulp tissue and thus encourage continued root development and apexogenesis.[140,335] Partial

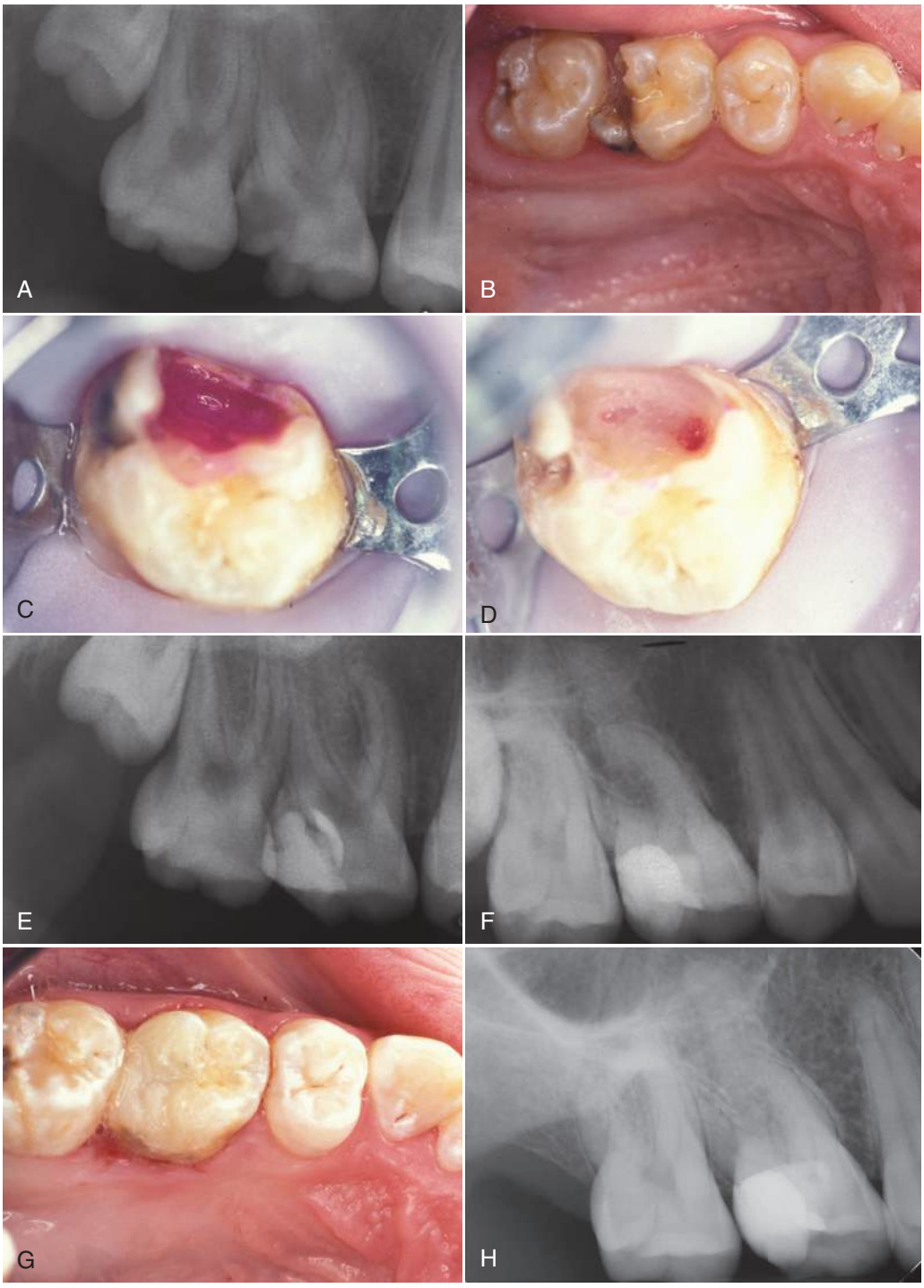


Fig. 24.12 A symptomatic 15-year-old patient presented for treatment with deep caries associated with the maxillary right first molar (#3). **A,** Preoperative radiograph showing deep caries but no evidence of apical pathosis. **B,** Clinical photograph of the carious molar. **C,** Caries removal using caries detector dye. **D,** Clinical view of pulp exposures after 5.25% sodium hypochlorite hemostasis. **E,** Periapical radiograph of direct mineral trioxide aggregate (MTA) pulp cap with wet cotton pellet and unbonded Photocore provisional. **F,** Permanent bonded composite placed 5 days after MTA direct pulp cap. **G,** Clinical photograph of final bonded composite restoration. **H,** Twenty-year 4-month recall radiograph without restorative intervention, patient was asymptomatic and responded normally to cold testing. (Reprinted courtesy Dr. George Bogen with permission.)

pulpotomy to treat direct pulp exposures in immature permanent teeth using CH products has been shown to be a reliable treatment option with proper case selection.[73,335] However, comparable success rates ranging from 83% to 100% have been demonstrated using CSCs for pulpotomies in both immature and mature permanent teeth.[23,25,39,242,313,361–363,398,399] Moreover, pulpotomies completed with MTA/CSCs do not show complications such as internal resorption, typically seen with CH, formocresol, and ferric sulfate pulpotomies (Fig. 24.16).[126,195,324]

Adult pulpotomy using CSCs is a promising treatment strategy of considerable interest. More important is the

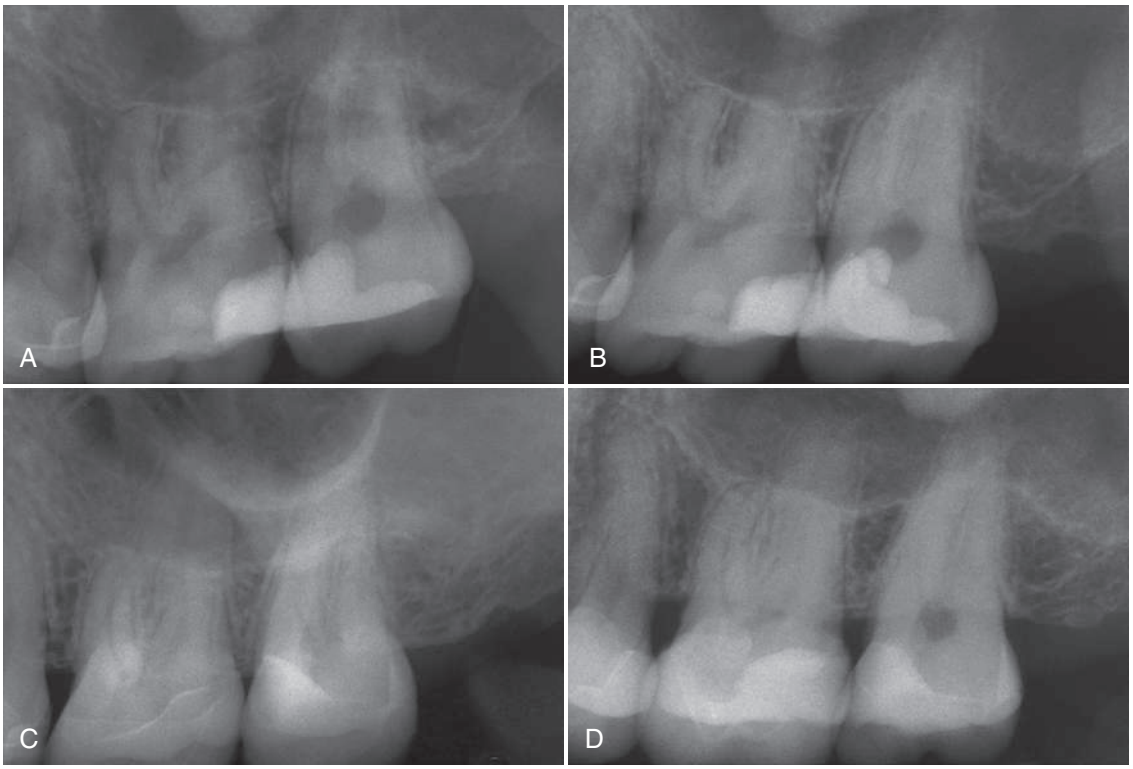


Fig. 24.13 A 23-year-old female patient presented with an IRM temporary restoration placed over a large pulp exposure after caries excavation of the maxillary left second molar (#15). The patient was referred for root canal treatment but had no history of pain and responded normally to sensibilty testing. **A,** Periapical radiograph showing temporary material in close proximity to mesial pulp horn. The tooth received a direct pulp cap with mineral trioxide aggregate (MTA) after further caries excavation and 5.25% sodium hypochlorite hemostasis. **B,** Radiograph showing direct MTA (Dentsply, Tulsa Dental, Tulsa, OK) pulp cap and bonded composite restoration placed at second visit. **C,** Two-year 5-month radiograph showing normal periapical structures and newly cemented porcelain onlay. **D,** Periapical radiographic review after 12 years and 8 months. The tooth responded positive to sensibility testing at both recall periods. (© Dr. George Bogen 2019.)

inclusion of symptomatic teeth in clinical investigations that exhibit irreversible pulpitis and show apical radiolucencies consistent with advanced pulp inflammation or partial pulp necrosis.[23,25,39,242,313,361–363,398,399] The potential for resolution of apical disease in teeth with inflamed pulps displaying apical rarefaction after pulpotomy procedures has been previously documented.[66,208] One multicentered randomized controlled trial complemented by an earlier study evaluated randomly allocated MTA and CEM adult pulpotomies at 2 and 5 years. The clinical outcomes for both materials were ≥98%, showing equal effectiveness for both CSCs in permanent molars associated with irreversible pulpitis and apical periodontitis.[22,24] Another investigation by the same group examined pulpotomy outcomes for CEM measured against conventional orthograde root canal therapy in mature molars diagnosed with irreversible pulpitis. It demonstrated comparable outcomes in pain reduction and the resolution of apical periodontitis, with continued tooth function and survival after 5 years (Fig. 24.17).[23] Both studies showed postoperative pain reduction and the resolution of apical pathosis in the majority of cases. Other investigations have confirmed significant pain reduction in symptomatic permanent teeth with deep caries using CSC pulpotomy.[152,361] Concurrent studies corroborate findings that pulpotomy treatments performed in mature permanent teeth with irreversible pulpitis and advanced caries using CSCs are a viable future treatment alternative.[22,23,25,242,361–363]

Indications for pulpotomy have recently transitioned from immature to mature teeth using CSCs, which may improve consumer acceptance and retention rates for patients seeking alternatives to conventional root canal therapy and protective ancillary care. Teeth that present with irreversible pulpitis and experience carious pulp exposures are normally treated with pulpectomy and orthograde root canal treatment coupled with restorative procedures designed to protect the tooth from fracture and microleakage. However, issues with cost effectiveness, holistic concerns, pain management, treatment time, and need for additional appointments when compared to orthograde root canal therapy are strong arguments in favor of this newer treatment option. Pulpotomy in permanent teeth using CSC materials can be recommended as a minimally invasive alternative to traditional root canal treatment.

Vital Pulp Therapy Techniques

DIAGNOSIS

A differential diagnosis based on clinical symptoms and radiographic findings is the aim in assessing pulp vitality. However, an accurate determination of the pulpal condition before treatment initiation is more challenging in younger patients.[356] Establishing a diagnosis of reversible versus irreversible pulpitis in immature teeth can be complicated by

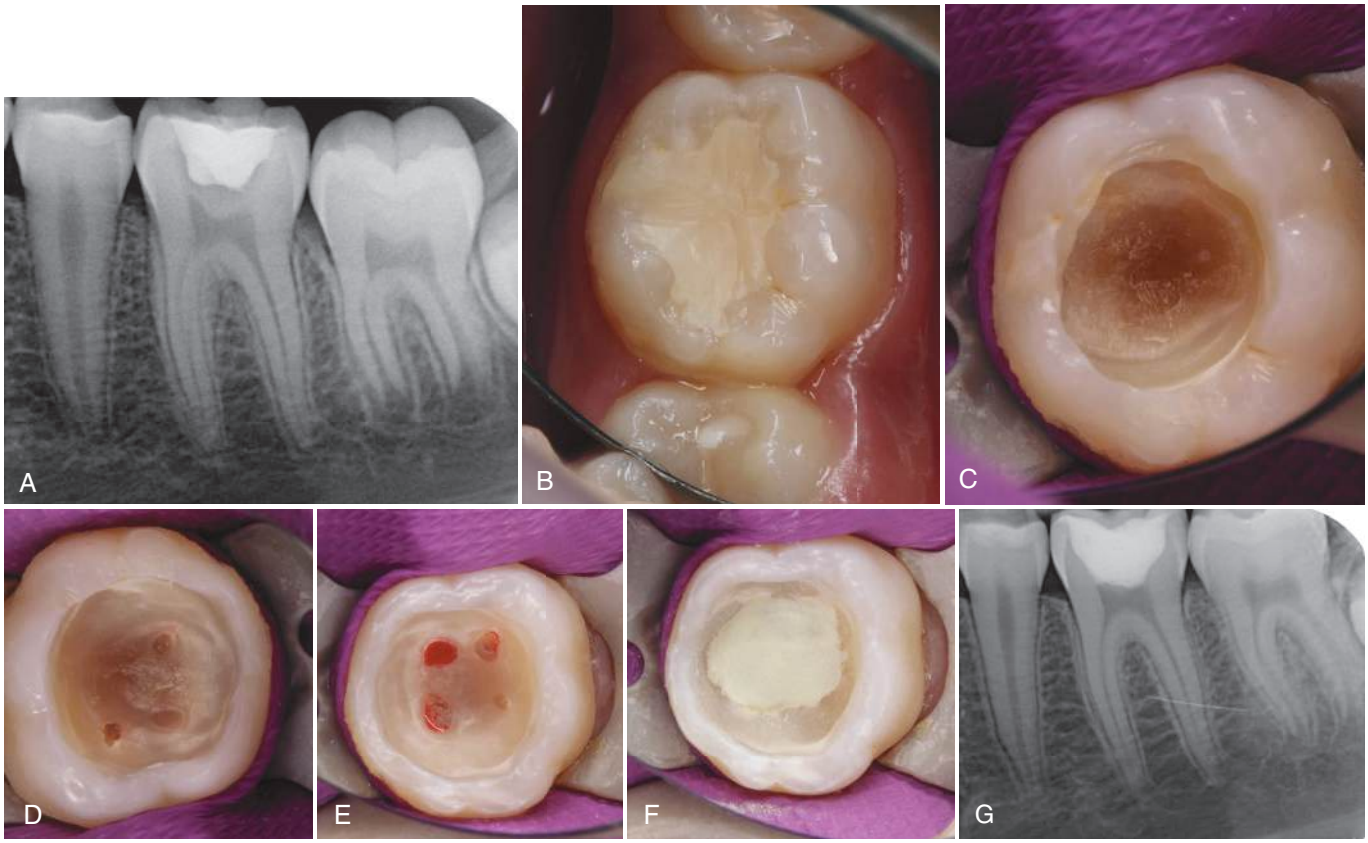


Fig. 24.14 A 12-year-old patient presented with a history of pain and cold sensitivity from a mandibular left first molar after having an occlusal composite restoration placed 16 weeks earlier. **A.** Periapical radiograph shows recurrent caries below base of the composite restoration. Note immature apices of the second molar. **B.** Clinical photograph of recently placed occlusal composite restoration. **C.** Photograph of initial caries removal showing remaining decay adjacent to pulpal roof. **D.** Photograph reveals three pulp horn exposures and presence of reparative dentin. **E.** Clinical view of three pulp horn exposures and hemostasis after application of 5% sodium hypochlorite. **F.** Photograph of bulk white mineral trioxide aggregate (MTA) (ProRoot MTA (Tulsa/Dentsply, Tulsa, OK) placement over exposures and surrounding dentin. The white MTA was covered with a flowable composite and the cavity restored with a bonded composite restoration using a two-bottle bonding system with enamel etching. **G,** One-year periapical radiograph showing normal apical structures. The tooth was asymptomatic and responded positively to vitality testing. (Courtesy Dr. Miguel Marques, Amsterdam. Netherlands.)

subjective symptoms and testing responses that may not accurately reflect pulp histopathology.[73] However, efforts should be directed toward the ultimate goal of pulpal preservation and continued apexogenesis in immature permanent teeth.[53,54] A diagnosis of irreversible pulpitis, based on signs and symptoms, along with clinical testing procedures, does not preclude vital pulp therapy options. Regardless of the treatment choice of pulp capping or partial or complete pulpotomy, preservation of the radicular pulp and apical papilla allows for root maturation in cases of trauma or deep caries.[146,147,169,342]

Diagnostic quality intraoral radiographs of the involved tooth must be taken to evaluate accurately the extent of root formation and periradicular or furcation changes associated with the periodontal ligament and supporting bone.[256] In young permanent teeth, the stage of root development directly influences the diagnosis and treatment options.[74] Because the faciolingual dimension of most immature roots is greater than the mesiodistal dimension, apical closure may be difficult to determine radiographically.[74,221] Teeth that demonstrate radiographic evidence of deep caries should not be planned for aggressive procedures, such as pulpectomy, without the benefit of thermal (cold) testing (see Fig. 24.1).

Before arriving at treatment decisions, the clinician should carefully assess all available information; the medical history, patient report, radiographic evidence, clinical evaluation, and vitality (cold) testing are recommended. Periodontal probing, mobility assessment, and the presence of any localized swelling or sinus tracts should be recorded during the evaluation. Radiographs, including bitewings and periapical views, should be evaluated for periapical and furcation pathosis, resorptive defects, and pulpal calcification resulting from trauma or previous restorations.

Patients with deep carious lesions often experience sensitivity to cold, heat, or sweet or acidic foods, and cold tests may evoke a short lingering response of 1 to 2 seconds. This may not be a definitive indicator that the pulp is irreversibly damaged. Determination of the pulpal condition with the aid of contemporary testing methods can be challenging, even for experienced clinicians, because of possible excessive responses to pulp percussion and palpation testing in children.[148,213,396] Clinical evidence indicates that cold testing with carbon dioxide ice is a more reliable prognosticator of pulp status in immature permanent teeth than electronic testing devices.[54] However, a diagnosis of irreversible pulpitis or pulp necrosis should be considered for teeth that generate pain on percussion.

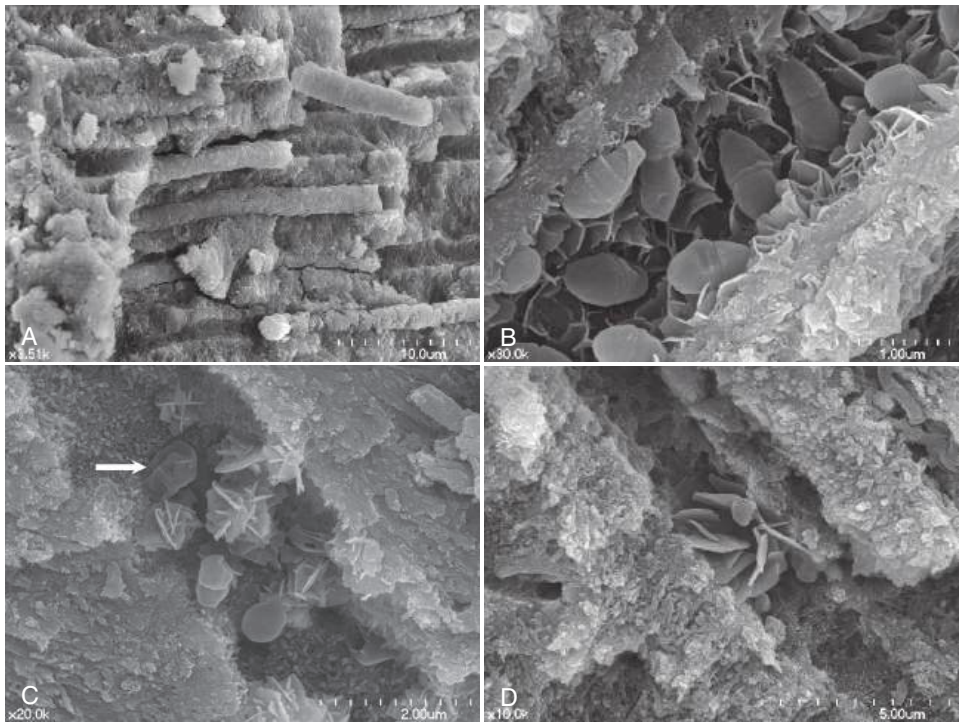


Fig. 24.15 Scanning electron micrographs of dentinal tubule obturation by calcium silicate cement (CSC) at the dentin/CSC interface after canals were seeded with *E. faecalis* in phosphate buffered saline (PBS). **A,** Tag formation in dentinal tubules at dentin interface in after application of OrthoMTA (BioMTA, Seoul, Korea) (×35,100). **B,** Four-week image showing *E. faecalis* entombed by growing crystalline structures within the dentinal tubule (×30,000). **C,** *White arrow* shows deformation and damaged cell membrane of *E. faecalis* within an aggregation of leaflet-like and needle-like crystals at 16 weeks (×20,000). **D,** Sixteen-week specimen showing complete blockage of the dentinal tubule lumen by CSC tight crystalline formation (×20,000). (Reprinted from Yoo JS, Chang SW, Oh SR, et al: Bacterial entombment by intratubular mineralization following orthograde mineral trioxide aggregate obturation: a scanning electron microscopy study. *Int J Oral Sci* 6:227, 2014, with permission.)

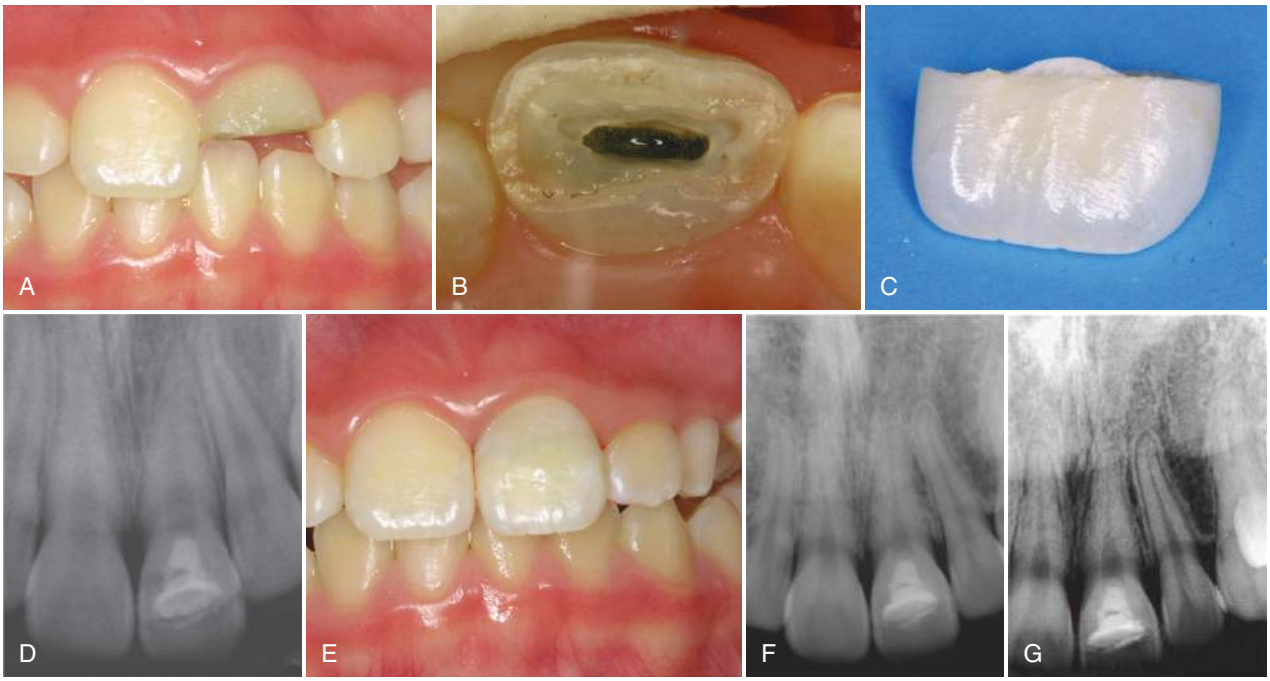


Fig. 24.16 Eleven-year-old female presented 3 hours after trauma with a horizontal crown fracture of the maxillary left central incisor. **A.** Clinical photograph showing midcoronal horizontal fracture. The pulp was exposed and exhibited normal vitality. **B,** Photograph of access cavity after Cvek pulpotomy and placement of gray mineral trioxide aggregate (MTA) (ProRoot MTA, Tulsa/Dentsply, Tulsa, OK). **C,** Photograph of fractured incisal segment before bonding and reattachment. **D,** Six-month radiograph with bonded fractured segment showing initial reparative bridge formation with absence of apical pathosis. The incisor was asymptomatic and responded normally to cold sensibility testing. **E,** Clinical photograph of incisor at 6-month recall. **F,** Eight-year radiographic review. **G,** Eleven-year radiographic recall. Tooth was asymptomatic wth normal mobility, probings, and positive response to cold test. (Courtesy Dr. Winston Chee and Dr. Stefan Zweig, Los Angeles, CA.)

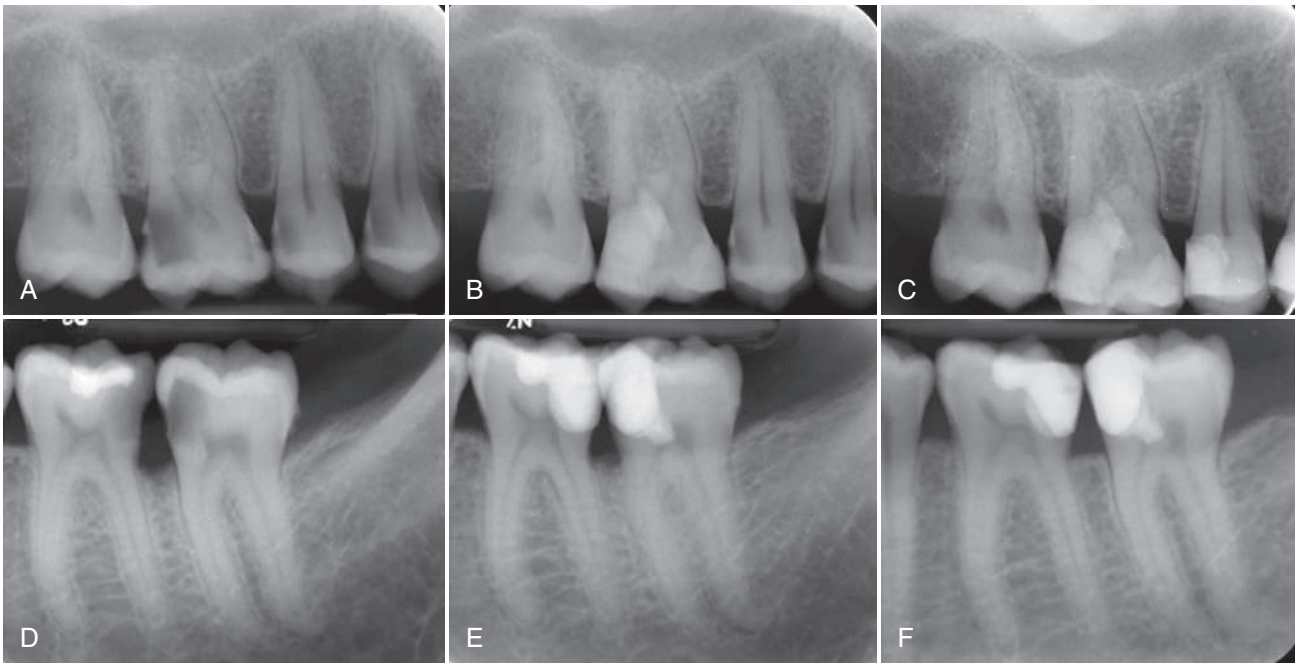


Fig. 24.17 Pulpotomy procedures completed for two patients clinically diagnosed with irreversible pulpitis using calcium enriched mixture (CEM) cement (BioniqueDent, Tehran, Iran). **A,** Preoperative radiograph from 27-year-old male patient with deep caries associated with maxillary right quadrant. The first molar was symptomatic with calcified tissue evident in the pulp chamber. **B,** Radiograph after complete pulpotomy using CEM cement restored with bonded composite. **C,** Five-year radiographic recall. The tooth was asymptomatic, firm, with normal probings. **D,** Preoperative radiograph showing deep caries in the mandibular left first (#19) and second (#18) molars in a 41-year-old male patient. Both teeth demonstrated lingering pain to cold testing and tooth #18 was sensitive to percussion. **E,** Postoperative radiograph showing completed miniature pulpotomy #19 and partial pulpotomy #18 after sodium hypochlorite hemostasis and application of CEM cement. Both molars were restored with bonded composites. Tooth #18 demonstrated resolution of symptoms after 24 hours. **F,** Six-year radiographic recall, patient remained asymptomatic and both teeth registered a positive response to sensibility testing. (Courtesy Dr. Saeed Asgary, Tehran, Iran.)

Clinically, the difference between reversible and irreversible pulpitis is often determined on the basis of the duration and intensity of pain.[147] Unprovoked, spontaneous pain of long duration or unrelenting symptoms forcing sleep deprivation are consistent with irreversible pulp inflammation or an acute periapical abscess.[50] The difficulty in predictably assessing pulpal health and potential degrees of inflammation has been recently addressed in an attempt to improve current terminology that may not accurately reflect pulp conditions during the inflammatory process.[401] It has been proposed that the terms *initial, mild, moderate,* and *severe pulpitis* be used to replace current terminology for reversible and irreversible pulpitis, better reflecting conditions encountered in the clinical setting. The recategorization of stages of pulpal inflammation attempts to better direct minimally invasive treatment selections using CSCs and is designed to improve outcomes for pulp tissue preservation. However, these assessments can be more complicated than previously proposed, as diagnosed pulp conditions are not always consistent with clinical findings during treatment.[321,401]

Another consideration in a patient with displacement trauma is transient apical breakdown, which mimics periapical radiolucencies.[16] Teeth that experience luxation-type injuries can discolor and may not respond to cold testing for up to 4 months before they recover normal color and vitality. Also, biologically or pharmacokinetically immunosuppressed patients may not respond to conventional treatments because of abnormal function of related repair mechanisms.[245,389] Most clinical investigations clearly indicate that successful outcomes for vital pulp therapy decrease as the patient's age increases. Although aging of the pulp diminishes pulpal volume, vascularity, and host immune responses, functional repair mechanisms can still provide favorable treatment outcomes in older patients (Fig. 24.18).[1,270]

The initial pulpal diagnosis can be confirmed after visualization of the exposed pulp and assessed during tissue hemostasis. If no hemorrhaging is seen, this area of the tissue is most likely necrotic and must be removed with a high-speed round diamond bur until bleeding is evident (Fig. 24.19). After hemostasis with NaOCl, a large bulk of CSC can be placed directly against the remaining tissue. Alternatively, if hemorrhage control cannot be achieved after 5 to 10 minutes of direct contact with 3% to 6% NaOCl, the pulp is likely to be irreversibly involved, and a full pulpotomy or pulpectomy is recommended.

Although the size of the pulp exposure has no significant bearing on the final outcome, some clinicians falsely assume that larger exposures have an unfavorable prognosis.[252] Pulp sizes are underestimated on radiographs.[82–84] The size of the pulp exposures may also be overestimated, which could affect the decision-making process, leading clinicians to abandon more conservative vital pulp treatment options.[163] Pulp dimensions may also vary between various racial groups and genders.[83,346,392]

Teeth that have a history of trauma or previous restorations or that display pulpal calcification have a poorer prognosis than teeth showing only initial caries. In the selection of a specific vital pulp treatment, it is important to consider

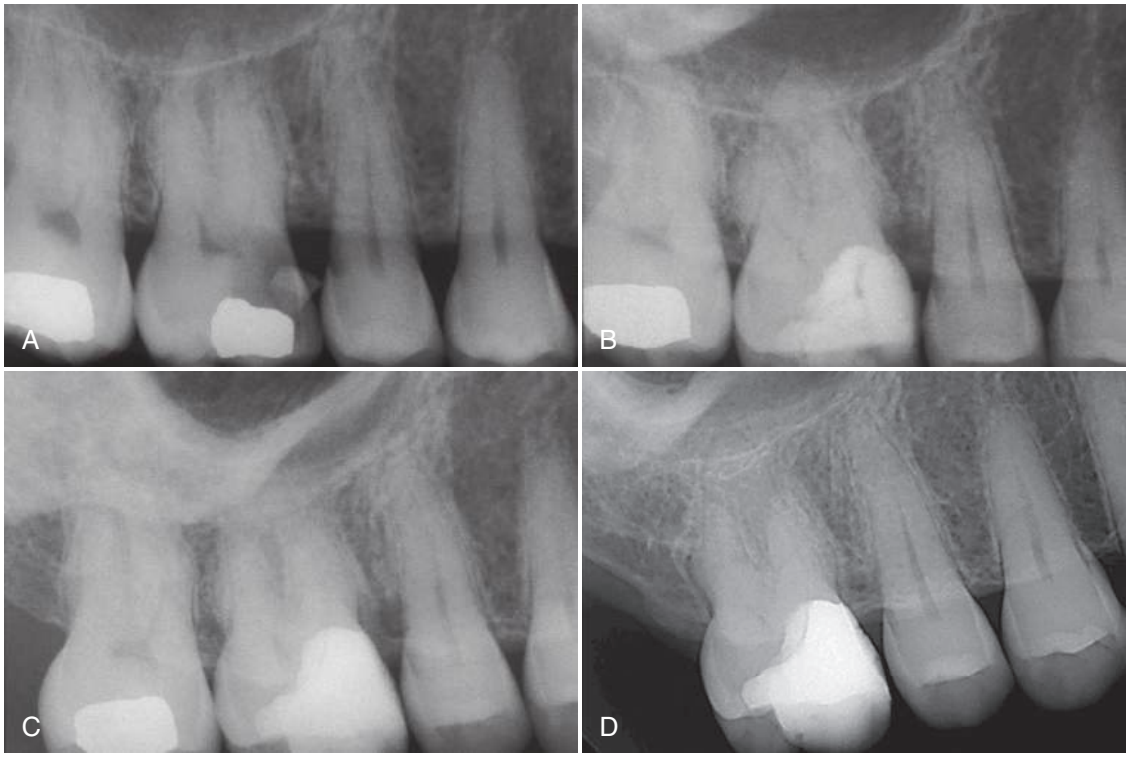


Fig. 24.18 A 51-year-old patient who presented with a deeply carious but asymptomatic maxillary right first molar. **A,** Preoperative radiograph reveals extensive mesial caries and an occlusal amalgam. **B,** Postoperative radiograph after 1.5 mm pulpal exposure, sodium hypochlorite hemostasis, mineral trioxide aggregate (ProRoot MTA, Tulsa/Dentsply, Tulsa, OK) direct pulp cap, wet cotton pellet placement, and Photocore provisionalization. **C,** Radiograph 1 week after pulp capping and placement of a permanent bonded composite restoration. The patient was asymptomatic and showed a positive response to cold testing. **D,** Six-year radiographic recall; cold testing revealed normal vitality. (© Dr. George Bogen.)

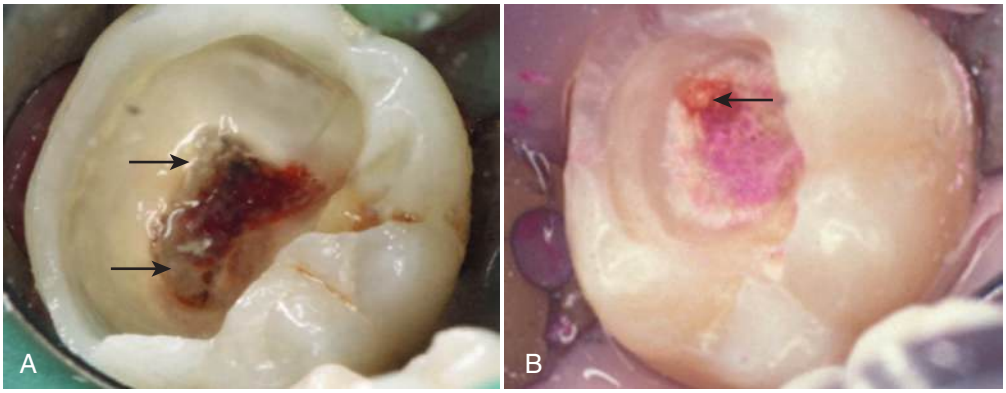


Fig. 24.19 Clinical examples of diseased pulp tissues after sodium hypochlorite hemostasis. **A,** Photograph of exposed pulp tissue of a mandibular right first molar in a 13-year-old patient. Note necrotic pulp tissue *(arrows)* that was subsequently removed with the remaining coronal pulp during a complete pulpotomy procedure. **B,** Clinical presentation of mandibular right molar in a 7-year-old patient after pulp exposure during caries excavation using a caries detector dye. Note the extruded seminecrotic, nonhemorrhagic tissue *(arrow)*. The tooth later underwent mineral trioxide aggregate partial pulpotomy and permanent restoration. (© Dr. George Bogen.)

the remaining tooth structure and future restorative plan. In patients with uncontrolled caries or extensive loss of coronal structure in which full coverage is indicated, pulpotomy rather than pulp capping is recommended. [60,73,373]

CARIES REMOVAL

The main objective in caries removal is the identification and complete removal of infected tissue while preserving sound tooth structure; this contributes to pulpal protection and continued vitality. Caries removal is enhanced with the aid of a caries detector dye and optical magnification; however, it must be understood that dyes can cause excessive and unnecessary removal of healthy tooth structure.[37,255,408]

Caries removal has traditionally been completed somewhat subjectively using hand instruments and slow-speed burs. The procedure is performed using an explorer and tactile sense to differentiate soft from hard dentin to determine infected from noninfected dental tissue. However, this method has shortcomings because clinicians may leave

decay at the dentinoenamel junction and unnecessarily remove dentin that has the potential to remineralize under a sealed restoration.[149] Furthermore, it has been found that the ability to remove caries varies among operators and during different time periods for the same operator.[149,150]

Investigators in the early 1970s used SEM to identify two different layers of carious dentin.[150] Teeth show two layers as the result of gram-positive bacteria releasing lactic acid as their main by-product. The outside carious layer subjacent to the enamel-dentin junction exhibited demineralized hydroxyapatite crystals that were dissolved by acidic bacterial by-products; this layer also featured unbound and altered collagen denatured by microbial proteolytic enzymes.[312] The second demineralized carious layer proximal to the pulp featured degraded hydroxyapatite crystals but contained collagen with intact intermolecular cross links unaffected by cariogenic acids.[149,150,331] If this deeper layer can be preserved during caries excavation, the remaining pulp tissue and odontoblasts will be subjected to less trauma, contributing to pulpal protection and survival. [282,326] This layer has a stronger capacity to remineralize when paired with bonded composite restorations to prevent bacterial microleakage.[269,282]

The two carious layers have been further classified into four zones when analyzed by atomic force microscopy and transverse digital microradiography.[312] Consistent with previous investigations, the four zones reinforce the concept that increasing levels of demineralization decrease the peritubular dentin rating and mechanical properties of dentin.[312]

Investigations in human, dog, and nonhuman primate models have demonstrated this regenerative characteristic in caries-affected dentin.[208,215,269,370,371] Several studies have questioned the efficacy of caries removal using a caries detector dye. Not all stainable dentin can be classified as infected, and the absence of staining does not eliminate the potential for residual cariogenic bacteria.[218,312] However, dyes allow the operator to visually inspect, with the aid of the DOM, infected dentin that may have been overlooked, particularly at the dentinoenamel junction, where it may compromise treatment outcome.[218,407] Although a compromise, it may be clinically preferable to inadvertently remove a small excess of dentin than to leave infected tissue.

HEMOSTATIC AGENTS

A wide range of hemostatic solutions and methods have been recommended to control a bleeding pulp exposure. These include various concentrations of NaOCl; 2% CHX; MTAD (Tulsa/Dentsply, Tulsa, OK); 30% hydrogen peroxide (Superoxol); ferric sulfate; disinfectants, such as Tubulicid (Global Dental Products, North Bellmore, NY); epinephrine; direct pressure with cotton pellets soaked in sterile water or saline; and the use of lasers.[8] NaOCl in concentrations of 1.5% to 5.25% is currently regarded as the most effective, safe, and inexpensive hemostatic solution for pulp capping and partial and complete pulpotomy procedures.[174,397] First used as a wound antiseptic during World War I and referred to as Dakin's solution, it was soon introduced for root canal irrigation.[100,104] NaOCl became a valuable hemostatic agent for direct pulp exposures in the late 1950s.[173,194,360] The antimicrobial solution provides hemostasis and disinfection of the dentin-pulp interface, chemical amputation of the blood clot and fibrin, biofilm removal, clearance of dentinal chips, and removal of damaged cells at the mechanical exposure site.[39,126,173,249] Concentrations of 1.5% to 5.25% in direct contact with pulp tissue do not appear to adversely alter pulp cell recruitment, cytodifferentiation, and hard tissue deposition.[121] Rinsing the exposed pulp tissue for 15 minutes with 5 mL NaOCl 5.25% (Clorox) leads to dissolution effects, but these effects are limited to the 3 to 5 upper cell layers without any changes in deeper pulp areas or the dentin. Perfused vital pulp tissue also buffers the chemical activity of NaOCl.[325]

An investigation into pulp-capped human third molars treated with either CH or a self-etching adhesive system examined pulps histologically at 30 and 60 days after the use of 2.5% NaOCl for hemostasis.[133] Histologic evidence demonstrated no impairment of the repair process after the use of NaOCl, although CH appeared to outperform the resin-based pulp capping agent. Current data clearly support the use of 1.25% to 5.25% NaOCl solutions in humans as a safe and appropriate hemostatic agent for DPC and pulpotomies.[5,34,121,133,173,192,252,384,388,397]

When pulp exposures occur in a carious field, the ability to attain hemostasis remains the most crucial factor in the success of vital pulp therapy. This was demonstrated in an innovative study that examined outcomes of teeth directly pulp capped with a hard setting CH after exposures were generated during caries excavation.[252] Caries removal during the investigation was aided by a caries detector dye with 10% NaOCl for hemostasis. The 2-year success rate was 81.8%. A statistical analysis of key factors revealed that preoperative thermal responses, percussion sensitivity, the diameter of the exposure, the age of the patient, and the tooth type and location had no significant influence on the outcome. The degree of bleeding and its control at the time of exposure constituted the most critical predictor of outcome. When hemostasis can be attained, pulpal repair and reparative dentin formation can proceed normally, in the absence of microbial challenges, when MTA and other CSCs are used as the pulp dressing.[87,210]

Sodium hypochlorite is an effective hemostatic agent and a valuable tool for assessing the difference between irreversibly and reversibly inflamed pulps as continued hemorrhaging after a 10-minute exposure strongly indicates irreversible pulpitis (Fig. 24.20).

TREATMENT CONSIDERATIONS

An important and often unrecognized aspect of DPC in a carious field is the potential survival of undetected bacteria in tubules adjacent to the exposure site even after caries removal. These microorganisms can compromise treatment even after meticulous excavation and disinfection. Unfortunately, traditional pulp capping protocols have directed clinicians to place pulp dressings only at the exposure site, without considering the adjacent dentin. It is therefore recommended that CSCs be placed over the exposure sites and onto the surrounding dentin to entomb residual microorganisms (see Fig. 24.15).[409,410]

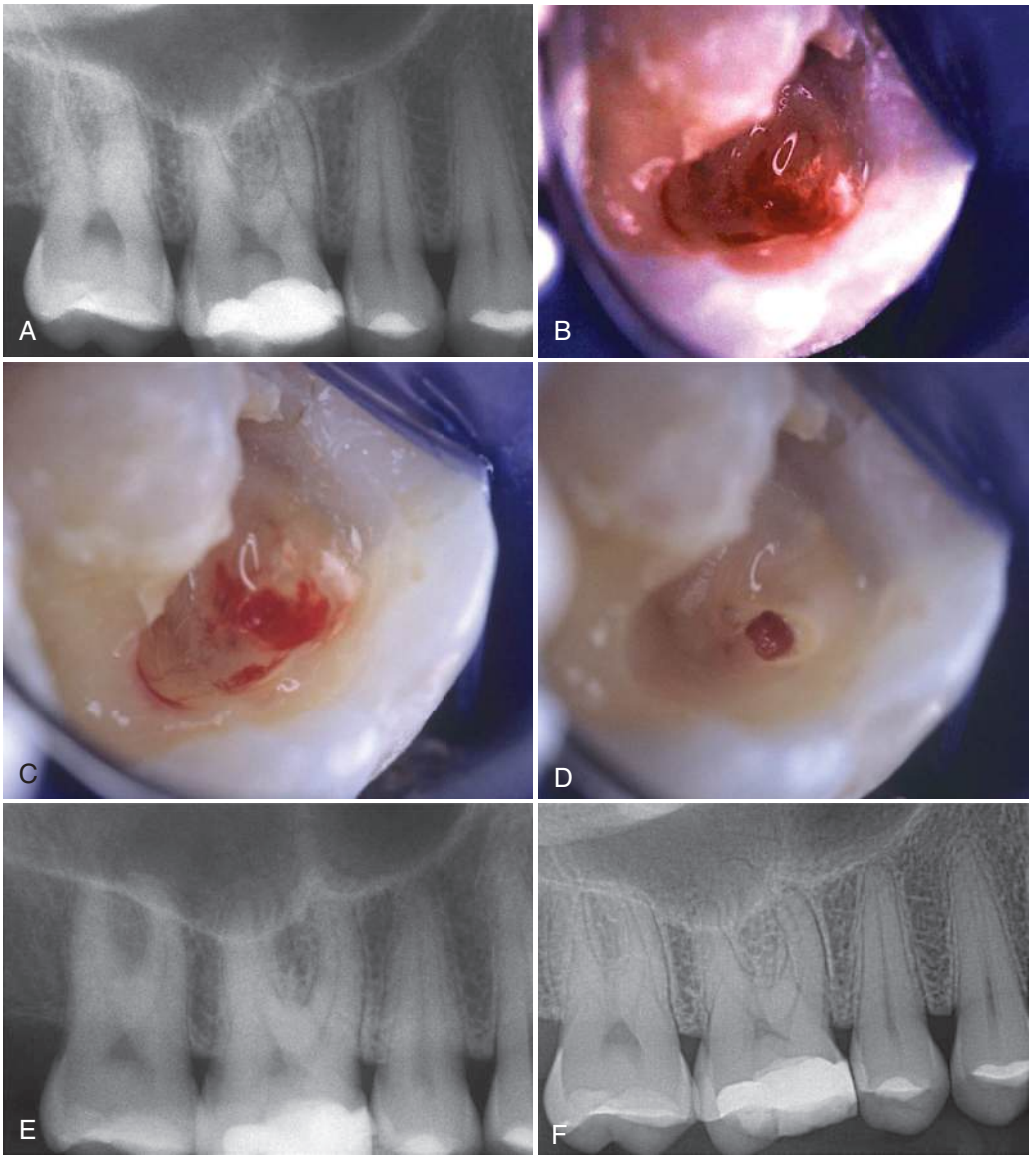


Fig. 24.20 Thirty-four-year-old female patient presented for root canal treatment with temporary restoration 1 week after caries excavation and pulp exposure of the maxillary right first molar (#3). The patient had mild discomfort and responded with a short lingering pain to cold testing. **A,** Periapical radiograph showing temporary restoration in tooth #3 with normal apical structures. **B,** Photograph of pulpal hemorrhage after temporary removal. **C,** Clinical photograph showing diminished bleeding after 5-minute exposure to 5.25% sodium hypochlorite. **D,** Photograph showing hemorrhage controlled in 10 minutes and 2 mm pulp exposure after continued caries excavation. The exposure and peripheral dentin received bulk OrthoMTA (BioMTA, Seoul, Korea) pulp capping and was covered with a thin wet cotton pellet and provisionalized with Photocore. The tooth was restored permanently with composite resin 1 week later. **E,** Two-year 6-month radiographic follow-up; tooth responded positive to sensibility testing. **F,** Control radiograph at 4 years. Patient exhibited normal pulp testing response and was asymptomatic with normal mobility and probings. (© Dr. George Bogen.)

The concept of biomineralization and bacterial entombment is particularly important when caries removal protocols are ineffective at completely removing undetected acidogenic microorganisms. During the CSC hydration process, calcium-deficient crystalline formation within the dentinal tubules produces bacterial entrapment, entombment, and neutralization.[409,414] This notable change in pulp capping strategy, based on CSC crystallization properties, can improve treatment outcomes in symptomatic and asymptomatic teeth featuring extensive caries. This includes cavities with multiple exposures after caries excavation or partial and complete pulptomies when indicated.[54] Together with an extensive bulk application of MTA/CSC to provide a cement thickness of 1.5 mm or greater, there is an increased likelihood of bacterial neutralization, minimizing further microbial challenges.

A disadvantage of CSCs is their long setting time of up to 228 minutes for ProRoot MTA and up to 85 minutes for Biodentine.[217] Thus according to the manufacturers' recommendation, CSCs should be allowed to set for a 15-minute minimum (e.g., Biodentine) or 2.5 hours (ProRoot MTA) before a permanent restoration can be placed. Nevertheless, in clinical use waiting even 15 minutes for set can lead to complications. To overcome these

clinical drawbacks and to bypass the long setting time, it has been suggested to cover the MTA or CSC with an RMGI, compomer, or flowable composite resin. The final restoration is then placed allowing the cement to set beneath the interlayer.[53,75]

Recently it was shown that 3 minutes after mixing MTA or Biodentine, it may be covered with a thin layer of a self-etching, self-bonding flowable composite resin (Vertise flow), an RGMI, or composite resin in combination with a self-etching dentin adhesive. All these restorative materials achieved shear bond strengths to Biodentine or MTA that were similar to those after 15 minutes and 2 days. A longer waiting time after mixing did not increase the adhesion of the lining materials to MTA or Biodentine.[333] Furthermore, the results of previous studies indicated that when placing RMGI on these cements, the time span between the mixing of CSCs and the application of RMGI did not influence the setting reaction of the cement. The moisture of the CSC had no impact on the setting reaction or setting time of RMGI or on the MTA-RMGI structural interface.[35,131,287] A final permanent restoration can therefore be provided at the same appointment.

Treatment Recommendations†

PULP CAPPING AND PULPOTOMY

1. After profound local anesthesia is achieved, the clinical crown is disinfected with NaOCl or CHX after dental dam isolation. Optical magnification and illumination (ideally with a DOM) are highly recommended. The cavity outline is completed using high-speed burs under constant water cooling.
2. Caries excavation is completed using long shank #2-6 round burs in a slow-speed handpiece augmented by hand instrumentation. Removal of caries may be guided by a detector dye if a DOM is used.
3. The cavity and exposure site(s) are immersed with 1.5% to 5.25% NaOCl. Heavy bleeding may be controlled with a cotton pellet moistened with NaOCl under moderate pressure for 5 minutes. If no bleeding is present after the pulp exposure, the area must be examined for necrotic tissue. This is removed with a high-speed round diamond bur under water coolant until bleeding tissue is exposed. Uncontrolled hemorrhaging after 10 minutes can be managed with a full pulpotomy or pulpectomy after assessing the radicular tissue condition. It is CRITICAL that all necrotic tissue is removed (see Fig. 24.19).
4. Prepare MTA or CSC according to the manufacturers' instructions.
5. Using a suitable instrument (composite instrument, Glick #1 endodontic instrument, or MTA carrier gun), apply a large bulk of CSC over the exposure side, including most of the surrounding dentin at a thickness of 1.5 to 3 mm.
6. Remove excess moisture at the site with a dry cotton pellet and leave a clear periphery of dentin for adhesive bonding. A total etch and rinse technique with a phosphoric acid gel is not possible at this stage because the MTA or CSC can be washed away. Either the CSC must show a firm set or a flowable composite/RMGI material must been accurately placed and completely cover the cement.
7. Apply a small amount of a light-curing flowable compomer, an RMGI (or an equivalent light-cured resin–glass ionomer liner), a self-etching, self-bonding flowable composite resin or a flowable composite resin in combination with a self-etching dentin adhesive to cover the CSC and light-cure according to the material's instructions. NOTE: Placement of flowable resins can be more challenging in cases of DPC than indirect pulp capping and pulpotomy procedures.
8. Etch the remaining cavity walls with 34% to 37% phosphoric acid gel for 60 seconds, rinse thoroughly and apply a dentin adhesive (fourth generation, two-bottle systems are highly recommended). Avoid using self-etching dentin adhesives (single-bottle systems).[290] Cure according to manufacturers' instructions.
9. Place a composite resin material incrementally as a permanent, final restoration. Light-cure according to manufacturer's instructions.
10. Pulp vitality should be assessed at the next appointment, preferably at 7 to 10 days using a cold test. Radiographic assessment is only needed in case of negative sensibility testing or onset of pain during the first year. Assessments can be made yearly or every 2 years, if indicated.

The Permanent Restoration

A definitive restoration should be provided during the pulp treatment appointment. Temporary restorations over MTA or CSC may lead to significantly lower success rates.[181,260] It has been reported that teeth which were permanently restored more than 2 days after DPC with MTA exhibited a diminished prognosis.[260] The negative impact of the time between DPC and placement of the permanent restoration was also shown for Biodentine and CH.[110,181] However, if the permanent restoration cannot be placed at the time of definitive pulp treatment due to management concerns, nonbonded translucent core materials such as Photocore (Kuraray Co. Ltd., Osaka, Japan) can provide acceptable protection for extended periods when used in conjunction with CSCs.[39]

The placement and quality of the permanent restoration can be crucial to the long-term maintenance of pulp vitality and may be more significant than the actual pulp treatment.[12,142,188,257,264] The aim of the final restoration is to complement the sealing ability of the pulp capping/pulpotomy material and defend the pulp from further microbial challenges. The selection of the restorative material, procedure execution, and adherence to proper restorative protocols can all contribute to minimizing potential restorative

†Adapted from Bogen and Chandler.[55]

microleakage. Microleakage around restorations may be more detrimental to vital pulp tissue than unfilled cavity preparations directly exposed to the oral environment.[330] An important consideration in both mature and immature permanent teeth is the conservation of the remaining tooth structure, which ultimately contributes to favorable long-term retention and function.[93,122,199,351] The incorporation of adhesive restorative materials as definitive final restorations minimizes tooth reduction, encourages anatomic preservation, and thus provides better pulpal protection and repair potential.[149,172]

Advances in dental materials have increased treatment options in the restoration of teeth that have undergone vital pulp procedures. Permanent restorations for pulp capped or pulpotomized permanent teeth can include composite resins, bonded or unbonded amalgam restorations, and cuspal coverage restorations. However, the more conservative the restorative treatment, the greater the probability of pulp survival.[149] Vital pulp therapy must be considered an injury to an already challenged connective tissue, and every effort should be directed toward minimizing further injurious procedures.[1] Factors that affect the repair mechanisms in pulp tissue can include the age of the patient, the depth and size of the cavity preparation, and the choice of restorative material.[283]

Total etch and self-etching systems produce excellent bond strengths to enamel, dentin, and cured CSCs.[27,288] The most durable bond strengths generated with contemporary materials are achieved using selective etching of the enamel with 34% to 37% phosphoric acid, followed by two-step, two-bottle, self-etching adhesive systems.[79,291,387] Current advances in adhesive technology have also shown that beneath the resin-interfused hybrid layer, an acid-base–resistant zone is formed that enhances the resistance of normal dentin to recurrent caries. This reinforced layer forms when monomer penetration and polymerization occur, and it demonstrates an ability to restrict (inhibit) primary and secondary caries.[290,291] As a result of their simplifications and improved bonding strengths, modern adhesives complement vital pulp therapy. Bonding procedures must strictly follow manufacturers' recommendations and be paired with dental dam placement.[79]

Postoperative Follow-Up and Recall

After treatment completion, the pulp status must be assessed periodically to ensure continued pulp vitality, normal function, and apical closure in immature teeth. Cold testing and radiographic evaluation most accurately assess continued pulp health and are excellent predicators for measuring survival rates. Recalls can address postoperative sensitivity, pulpal degeneration or necrosis, and indications for further endodontic and restorative care. Review also allows detection of emerging complications such as recurrent caries, poor hygiene, restoration failures, and cuspal fractures.

Patient compliance for recall in asymptomatic cases can be challenging. Because some parents do not practice regular preventive care and may not have a strong background in basic oral health care, recall compliance rates for children can be unpredictable.[205,311] Although recall has been traditionally based on a 6-month checkup and oral prophylaxis period, this has come into question.[265] Recalls can be tailored individually, based on patient need, symptomology, caries indices, periodontal status, and craniofacial development assessment in younger patients.[292,391]

A tentative diagnosis of tooth survivability can be made at the 3-month recall.[252] One study demonstrated that the prognosis for long-term pulp survival can be established at an observation period of 21 to 24 months.[252]

Radiographic evidence of root-end closure in immature adult teeth is a reliable prognostic marker of continued pulp vitality (Fig. 24.21).[36] Apexogenesis after DPC or pulpotomy procedures using CSCs should proceed normally in healthy patients at a progressive rate.[39,305,394,400] The observed root maturation should also follow a predictable pattern of tooth development that coincides with the contralateral teeth in the same patient when the teeth are compared radiographically and viewed chronologically.[36]

Observation of contralateral tooth development can be an invaluable method of measuring the success of vital pulp therapy when using CSCs. Alternatively by comparison, in traumatized teeth that exhibit necrotic pulps and periradicular pathosis, CSCs can be used as an apical plug to stimulate the apical papilla to promote barrier formation, which may require 5 to 20 months for completion (apexification).[7,90,344] If MTA or another CSC is substituted for CH in vital pulp therapy procedures or apexification, similar time periods for apical maturation or barrier formation can be anticipated (Fig. 24.22).[132,259 261]

In the absence of microbial challenges, the human dental pulp demonstrates an exceptional regenerative capacity when treated with CSCs independent of patient age. Together with our advances in pulp biology and dental materials, changes in treatment protocols that encourage dental pulp preservation and survival will ultimately contribute to improved dental health for all patients requiring vital pulp therapy.

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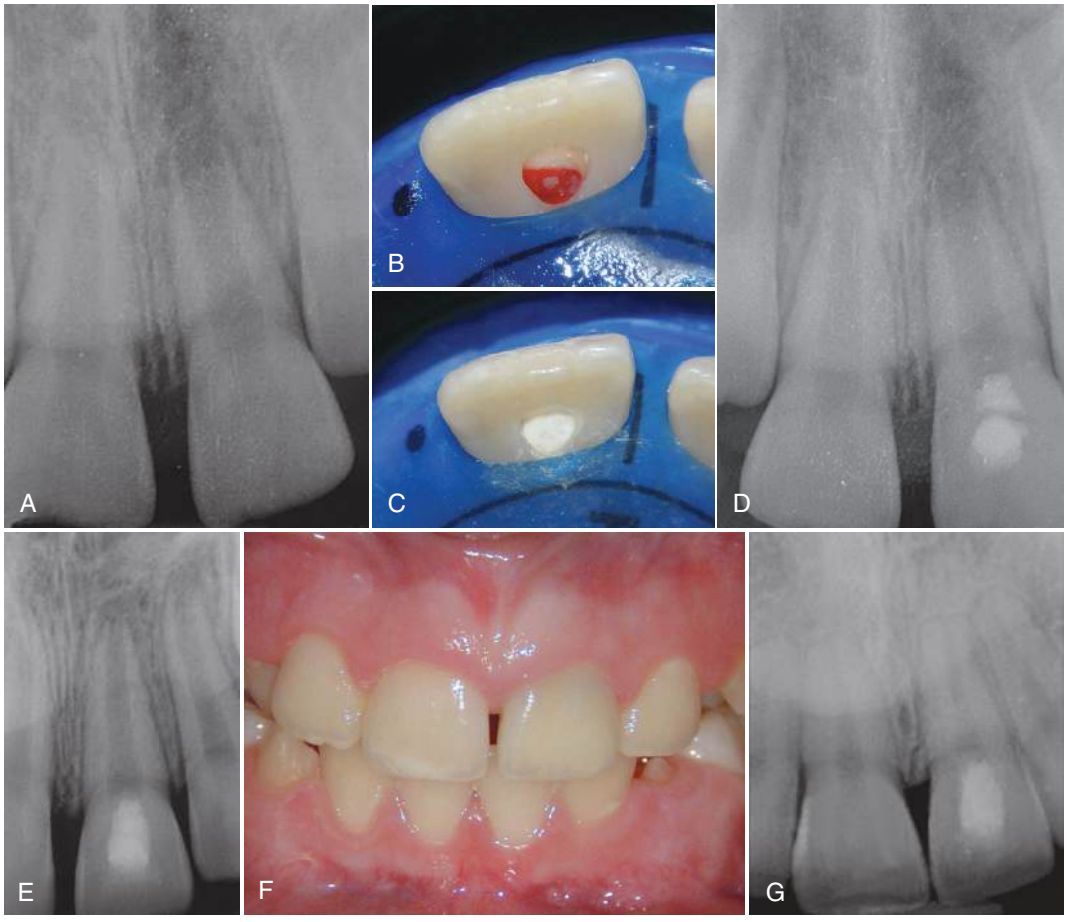


Fig. 24.21 An 8-year-old male presented following a traumatic injury to the maxillary left central incisor 2 months earlier. **A,** Periapical radiograph revealed an open apex with a developing radiolucency. The tooth was sensitive to palpation, percussion and negative to cold testing. The diagnosis was necrotic pulp and acute apical periodontitis. **B,** Clinical photograph after access cavity preparation, vital pulp tissue visible coronally. The diagnosis was changed to reversible pulpitis and a partial (Cvek) pulpotomy was performed. **C,** Photograph of white mineral trioxide aggregate (MTA) (ProRoot MTA, Tulsa/Dentsply, Tulsa, OK) applied over the pulp to a thickness of approximately 3 mm. **D,** Radiograph after MTA pulpotomy with provisional restoration. A definitive composite restoration was placed 1 week later. **E,** One-year radiographic recall showing advancing apical closure. **F,** Four-year clinical photograph reveals a slight grey discoloration from the MTA in the cervical area of the tooth. Parents declined option of internal bleaching. **G,** Radiographic review at 9 years reveals apical maturation and thickening of the root walls. The tooth responded positively to cold sensibility testing. (Reprinted courtesy Dr. Marga H. Ree, Purmerend, Netherlands with permission.)

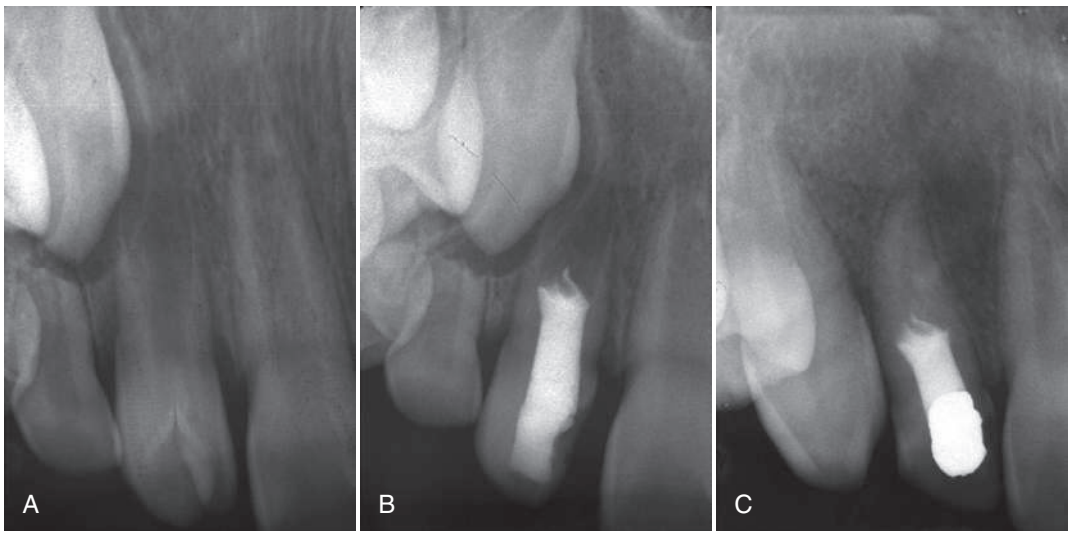


Fig. 24.22 **A,** Preoperative radiograph of a 7-year-old male patient exhibiting a dens evaginatus anatomical anomaly associated with the maxillary right lateral incisor. The parents had been advised to extract the tooth, which was symptomatic and nonresponsive to vitality tests. **B,** A periapical radiograph after cleaning the coronal portion of the canal and irrigation with 2.12% sodium hypochlorite. A coronal plug using ProRoot MTA (Tulsa/Dentsply, Tulsa, OK) was placed and sealed with a provisional restoration. **C,** Four-year radiographic review demonstrates complete root formation with maturation and apical root closure. The tooth was asymptomatic with normal mobility and probings. (Courtesy Dr. Mahmoud Torabinejad.)

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